

# INORGANIC CHEMISTRY

PERIODIC PROPERTIES\_MICRO\_XI



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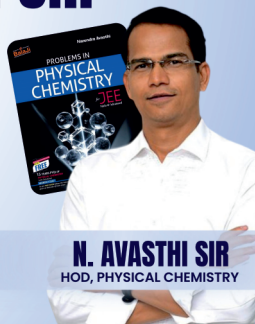
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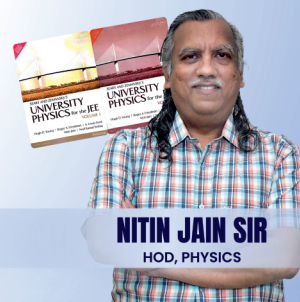


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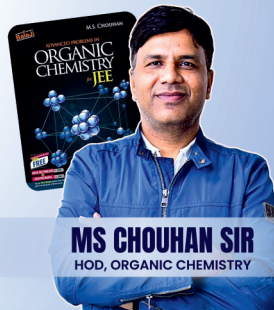


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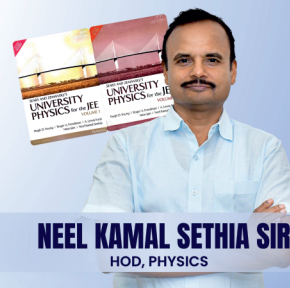


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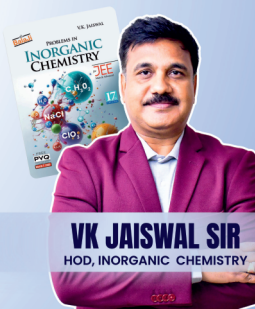


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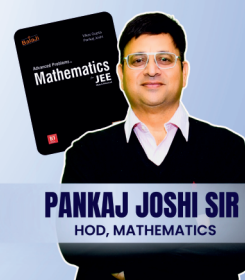


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- Co-founder, Director, Vibrant Academy
- Teaching experience of 24 years
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# INORGANIC CHEMISTRY

## TARGET IIT JEE MICRO-XI

# CHEMICAL BONDING

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# EXERCISE - I

## Causes of chemical combination :

- When two atoms combine to form a molecules:  
(1) Energy is released (2) Energy is absorbed  
(3) Energy is neither released nor absorbed (4) Energy may either released or aborbed
- The combination of atoms occur because they want :  
(1) To decrease number of electrons in the outer most orbit  
(2) To attain an inert gas configuration  
(3) To increase number of electrons in the outer most orbit  
(4) To attain 18 electrons in the outermost orbit
- Which condition favours the bond formation:  
(1) Maximum attraction and maximum potential energy  
(2) Minimum attraction and minimum potential energy  
(3) Minimum potential energy and maximum attraction  
(4) None of the above

## Ionic bond :

- An electrovalent bond or ionic bond is formed between:  
(1) Two electronegative atoms (2) Two metals  
(3) Electropositive and electronegative atoms (4) Two electropositive atoms
- Most favourable conditions for electrovalent bonding are:  
(1) Low ionisation potential of one atom and high electron affinity of the other atom  
(2) High electron affinity and high ionisation potential of both the atoms  
(3) Low electron affinity and low ionisation potential of both the atoms  
(4) High ionisation potential of one atom and low electrons afinity of the other atom
- The electronegativity of Cesium is 0.7 and fluorine is 4.0. The bond formed between the two is:  
(1) Covalent (2) Electrovalent/ionic (3) Coordinate (4) Metallic
- Electrovalent bond or ionic bond is formed by:  
(1) Sharing of electons (2) Donaton of electrons  
(3) Transfer of electrons (4) None of these
- Element X is strongly electropositive and is strongly electronegative. Both are univalent. The compound formed would be:  
(1)  $X^+ Y^-$  (2)  $X - Y$  (3)  $X^- Y^+$  (4)  $X \rightarrow Y$
- As compared to covalent compounds electrovalent compounds generally possess:  
(1) High m.p. and high b.p. (2) Low m.p. and low b.p.  
(3) Low m.p. and high b.p. (4) high m.p. and low b.p.
- Which has the highest lattice energy?  
(1) LiF (2) LiCl (3) NaCl (4) MgO
- Solid NaCl is a bad conductor of electricity because:  
(1) In solid NaCl there are no ions (2) Solid NaCl is covalent  
(3) In solid NaCl there is no mobility of ions (4) In solid NaCl there are no electrons

12. Electrovalent compounds or ionic compounds do not show isomerism. The reason is:  
 (1) Presence of ions (2) Strong electrostatic force of attraction  
 (3) Brittleness (4) Non-directional nature of ionic bond
13. Molten sodium chloride conducts electricity due to the presence of :  
 (1) Free electrons (2) Free molecules  
 (3) Free sodium and chloride atoms (4) Free sodium and chloride ions
14. Choose **incorrect** statement about ionic compounds :  
 (1) An ionic solid does not conduct electricity  
 (2) They are more soluble in polar solvents like water  
 (3) Their enthalpy of formation ( $\Delta H_f$ ) is positive  
 (4) They generally have higher melting point than covalent compounds

#### Covalent Bond :

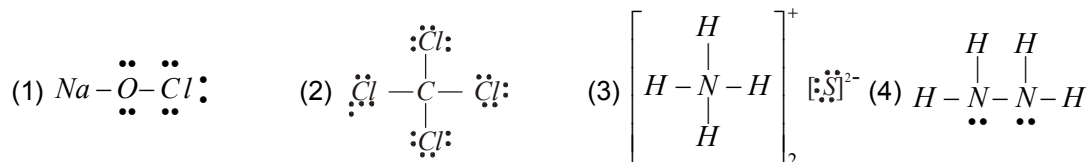
15. Which of the following statement is not true about covalent compounds:  
 (1) They may exhibit space isomerism (2) They have low melting and boiling points  
 (3) They show ionic reactions (4) They show molecular reactions
16. In a triple bond there is sharing of:  
 (1) 3 electrons (2) 4 electrons (3) Several electrons (4) 6 electrons
17. Which of the following bonds will have directional character:  
 (1) Ionic bond (2) Metallic bond (3) Covalent bond (4) None
18. The type of bond formed between two electronegative atoms would be:  
 (1) Covalent (2) Ionic (3) Coordinate (4) All the above
19. Elements have electronegativities 2.5 and 4.0, bond formed between them would be:  
 (1) Ionic (2) Covalent (3) Co-ordinate (4) Metallic
20. Which of the following is not a characteristic of a covalent compound:  
 (1) It has low melting point and boiling point  
 (2) Covalent bond is formed between two atoms having very small electronegativity difference  
 (3) They have no definite geometry  
 (4) They are generally insoluble in water
21.  $\text{H}_3\text{BO}_3$  is  
 (1) monobasic weak Lewis acid (2) monobasic weak Bronsted acid  
 (3) monobasic strong Lewis acid (4) tribasic and weak Bronsted acid
22. Which one of the following is smallest in size ?  
 (1)  $\text{N}^{3-}$  (2)  $\text{O}^{2-}$  (3)  $\text{F}^-$  (4)  $\text{Na}^+$

#### Coordinate Bond :

23. In Co-ordinate bond, the acceptor atoms must essentially contain in its valency shell an orbital:  
 (1) With paired electron (2) With single electron  
 (3) With no electron (4) With three electron
24. Correct statement regarding this reaction  

$$\text{BF}_3 + \text{NH}_3 \longrightarrow [\text{F}_3\text{B} \leftarrow \text{NH}_3]$$
 (1) Hybridisation of N is changed (2) Hybridisation of B is changed  
 (3)  $\text{NH}_3$  act as a lewis base (4) (2) & (3) both

25. The correct statement for the reaction :  $\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$
- (1) Hybridisation state is changed (2) Bond angle increases  
(3)  $\text{NH}_3$  act as a Lewis acid (4) Electron geometry is changed
26. The bonds present in  $\text{N}_2\text{O}_5$  are:
- (1) Only ionic (2) Covalent & coordinate  
(3) Only covalent (4) Covalent & ionic
27. The pair of compounds which can form a co-ordinate bond is:
- (1)  $(\text{CH}_3)_3\text{B}$  and  $(\text{CH}_3)_3\text{N}$  (2)  $\text{HCl}$  and  $\text{HBr}$   
(3)  $\text{BF}_3$  and  $\text{NH}_3$  (4) (1) and (3) both
28. In which compound coordinate bond is present:
- (1)  $\text{NH}_3$  (2)  $\text{NH}_4\text{OH}$  (3)  $\text{H}_2\text{O}$  (4)  $\text{HCl}$
29. Which of the following molecules does not have coordinate bonds?
- (1)  $\text{H}_3\text{O}^+$  (2)  $\text{NO}_3^-$  (3)  $\text{O}_3$  (4)  $\text{CO}_3^{2-}$
30. Which of the following Lewis diagrams is(are) incorrect ?

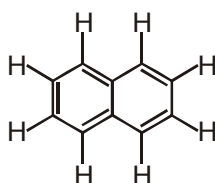


### Valence Bond Theory (VBT) :

31. A sigma bond is formed by the overlapping of :
- (1) s-s orbital alone  
(2) s and p orbitals alone  
(3) s-s, s-p or p-p orbitals along internuclear axis  
(4) p-p orbital along the sides
32. Which overlapping is involved in  $\text{HCl}$  molecule:
- (1) s-s overlap (2) p-p overlap (3) s-d overlap (4) s-p overlap
33. which of the following configuration shows second excitation state of Iodine:
- (1)  $\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$  (2)  $\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$   
(3)  $\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$  (4)  $\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$
34. Which of the following compound is formed in the second excitation state of sulphur atom:
- (1)  $\text{SF}_4$  (2)  $\text{SF}_6$  (3)  $\text{SF}_2$  (4) None
35. The strength of bonds by 2s – 2s, 2p – 2p and 2p – 2s overlapping has the order:
- (1) s – s > p – p > s – p (2) s – s > p – s > p – p  
(3) p – p > s – p > s – s (4) p – p > s – s > p – s
36. The triple bond in ethyne is made up of:
- (1) Three sigma bonds (2) Three  $\pi$  - bonds  
(3) One sigma and two  $\pi$  bonds (4) Two sigma and one  $\pi$  bond

37. In which of the excitation state of chlorine  $\text{ClF}_3$  is formed:  
 (1) In ground state (2) In third excitation state  
 (3) In first excitation state (4) In second excitation state
38. Variable covalency is exhibited by:  
 (1) P and S (2) N and O (3) N and P (4) F and Cl
39. Which is not characteristic of  $\pi$ -bond :  
 (1)  $\pi$ -bond is formed when a sigma bond already formed  
 (2)  $\pi$ -bond are formed from hybrid orbitals  
 (3)  $\pi$ - bond may be formed by the overlapping of p-orbitals  
 (4)  $\pi$ - bond results from lateral overlap of atomic orbital
40.  $\pi$  bond is formed :  
 (1) By overlapping of hybridised orbital (2) Overlapping of axial p-p orbitals  
 (3) Head on overlapping of s – s orbitals (4) By p – p lateral overlapping
41. Number of  $\sigma$  and  $\pi$  bonds present in  
 $\text{CH}_3 - \text{CH} = \text{CH} - \text{C} \equiv \text{CH}$  are:  
 (1)  $10\sigma, 3\pi$  (2)  $10\sigma, 2\pi$  (3)  $9\sigma, 2\pi$  (4)  $8\sigma, 3\pi$
42. Which of the following fluorides does not exists?  
 (1)  $\text{NF}_5$  (2)  $\text{PF}_5$  (3)  $\text{AsF}_5$  (4)  $\text{SbF}_5$
43. p-p overlapping will be observed in the molecules of -  
 (1)  $\text{H}_2$  (2)  $\text{HBr}$  (3)  $\text{HCl}$  (4)  $\text{Cl}_2$
44. Nitrogen does not form  $\text{NF}_5$  because:  
 (1) Nitrogen is member of V group  
 (2) It contains no empty d-orbital  
 (3) The bond energy of  $\text{N} \equiv \text{N}$  is very high  
 (4) Inert pair effect exists in the molecule
45. The correct order of bond length is  
 (1)  $\text{C} - \text{C} < \text{C} = \text{C} < \text{C} \equiv \text{C}$  (2)  $\text{C} \equiv \text{C} < \text{C} = \text{C} < \text{C} - \text{C}$   
 (3)  $\text{C} = \text{C} < \text{C} \equiv \text{C} < \text{C} - \text{C}$  (4)  $\text{C} = \text{C} < \text{C} - \text{C} < \text{C} \equiv \text{C}$
46. Which of the following statements is incorrect ?  
 (1) Double bonds is shorter than a single bond  
 (2)  $\sigma$  – bond is weaker than  $\pi$  bond  
 (3) Double bond is stronger than a single bond  
 (4) Covalent bond is stronger than a hydrogen bond
47. Fluorine molecule is formed by:  
 (1) the axial p-p orbital overlap (2) the sideways p-p orbital overlap  
 (3) the s-s orbital overlap (4) the s-p orbital overlap
48. Consider the following statements:  
 I. A sigma ( $\sigma$ ) bond is formed when two s-orbitals overlap  
 II. A pi( $\pi$ ) bond is formed when two p-orbitals overlap axially  
 III. A  $\sigma$  - bond is weaker than  $\pi$  -bond  
 Which of the above statements is/are correct?  
 (1) I and II (2) II and III (3) I alone (4) II alone

49. Number and type of bonds between two carbon atoms in  $\text{CaC}_2$  are :  
 (1) one sigma ( $\sigma$ ) and one pi ( $\pi$ ) bond (2) one  $\sigma$  and two  $\pi$  bonds  
 (3) one  $\sigma$  and one and a half  $\pi$  bond (4) one  $\sigma$  bond
50. How many  $\sigma$ - and  $\pi$ - bonds are there in salicylic acid (2-hydroxy benzoic acid)?  
 (1)  $10\sigma, 4\pi$  (2)  $16\sigma, 4\pi$  (3)  $18\sigma, 2\pi$  (4)  $16\sigma, 2\pi$
51. Strongest bond is formed by the head on overlapping of :  
 (1) 2s- and 2p- orbitals (2) 2p- and 2p- orbitals  
 (3) 2s- and 2s- orbitals (4) All
52. The ratio of  $\sigma$  and  $\pi$  bonds in benzene is :  
 (1) 2 (2) 6 (3) 4 (4) 8
53. Number of  $\pi$ -bonds and  $\sigma$ -bonds in the following structure is



- (1) 6, 19 (2) 4, 20 (3) 5, 19 (4) 5, 20

#### Hybridisation :

54. In the protonation of  $\text{H}_2\text{O}$ , change occurs in :  
 (1) Hybridisation state of oxygen (2) Shape of molecule  
 (3) Hybridisation and shape both (4) None
55. The d-orbitals involved in  $\text{sp}^3\text{d}$  hybridisation is :  
 (1)  $d_{x^2-y^2}$  (2)  $d_{z^2}$  (3)  $d_{xy}$  (4)  $d_{xz}$
56. A  $\text{sp}^3$  hybrid orbital contains:  
 (1)  $\frac{3}{4}$  s-character (2)  $\frac{1}{4}$  p-character (3)  $\frac{3}{4}$  p-character (4)  $\frac{1}{2}$  s-character
57. In the compound  ${}^1\text{CH}_2 = {}^2\text{CH} - {}^3\text{CH}_2 - {}^4\text{CH}_2 - {}^5\text{C} \equiv {}^6\text{CH}$ , the  $\text{C}_2 - \text{C}_3$  bond formed by the overlapping of :  
 (1)  $\text{sp} - \text{sp}^2$  (2)  $\text{sp}^3 - \text{sp}^3$  (3)  $\text{sp} - \text{sp}^3$  (4)  $\text{sp}^2 - \text{sp}^3$
58. Which of the set of species have same hybridisation state but different shapes:  
 (1)  $\text{NO}_2^+$ ,  $\text{NO}_2$ ,  $\text{NO}_2^-$  (2)  $\text{ClO}_4^-$ ,  $\text{SF}_4$ ,  $\text{XeF}_4$  (3)  $\text{NH}_4^+$ ,  $\text{H}_3\text{O}^+$ ,  $\text{OF}_2$  (4)  $\text{SO}_4^{2-}$ ,  $\text{PO}_4^{3-}$ ,  $\text{ClO}_4^-$
59. Which of the following elements can not exhibit  $\text{sp}^3\text{d}$  hybridisation state:  
 (1) C (2) P (3) Cl (4) B  
 Correct answer is:  
 (1) 1, 3 (2) 1, 4 (3) 2, 3 (4) 2, 4

60. In the protonation of  $\text{NH}_3$  molecule, following statement is true:  
 (1) A covalent bond is formed (2) Hydrogen bond is formed  
 (3) Hybridisation state of N is changed (4) Shape of  $\text{NH}_3$  molecule is changed
61. The shape of sulphate ion is:  
 (1) Hexagonal (2) Square planar (3) Trigonal bipyramidal (4) Tetrahedral
62. In which following set of compound/ion has linear geometry:  
 (1)  $\text{CH}_4$ ,  $\text{NH}_4^+$ ,  $\text{BH}_4^-$  (2)  $\text{CO}_3^{2-}$ ,  $\text{NO}_3^-$ ,  $\text{BF}_3$  (3)  $\text{NO}_2^+$ ,  $\text{CO}_2$ ,  $\text{N}_3^-$  (4)  $\text{BeCl}_2$ ,  $\text{BCl}_3$ ,  $\text{CH}_4$
63. Which of the molecule is trigonal bipyramidal :  
 (1)  $\text{BF}_3$  (2)  $\text{CH}_4$  (3)  $\text{PCl}_5$  (4)  $\text{SF}_6$
64. The type of hybrid orbitals used by chlorine atom in  $\text{ClO}^-$ ,  $\text{ClO}_2^-$ ,  $\text{ClO}_3^-$  and  $\text{ClO}_4^-$  is/are:  
 (1) sp,  $\text{sp}^2$ ,  $\text{sp}^3$  and  $\text{sp}^3\text{d}$  (2) sp and  $\text{sp}^3$   
 (3) Only  $\text{sp}^3$  (4) Only sp
65. Which of the following having a square planar structure is:  
 (1)  $\text{NH}_4^+$  (2)  $\text{BF}_4^-$  (3)  $\text{XeF}_4$  (4)  $\text{CCl}_4$
66. The shape of  $\text{I}_3^-$  is :  
 (1) Tetrahedral (2) Linear (3) T-shape (4) Trigonal
67. Among the following compounds the one that is polar and has the central atom with  $\text{sp}^3$  hybridisation is:  
 (1)  $\text{H}_2\text{CO}_3$  (2)  $\text{SiF}_4$  (3)  $\text{BF}_3$  (4)  $\text{HClO}_2$
68. Which of the following will be octahedral:  
 (1)  $\text{SF}_6$  (2)  $\text{BF}_4^-$  (3)  $\text{PCl}_5$  (4)  $\text{XeF}_6$
69. The shape of methyl cation ( $\text{CH}_3^+$ ) is likely to be:  
 (1) linear (2) pyramidal (3) planar (4) spherical
70. The structure of  $\text{XeF}_2$  involves hybridization of the type :  
 (1)  $\text{sp}^3$  (2)  $\text{sp}^3\text{d}^3$  (3)  $\text{sp}^3\text{d}$  (4)  $\text{sp}^3\text{d}^2$
71. In the  $\text{XeF}_4$  molecule, the Xe atom is in the  
 (1)  $\text{sp}^2$ -hybridized state (2)  $\text{sp}^3$ -hybridised state  
 (3)  $\text{sp}^3\text{d}$ -hybridized state (4)  $\text{sp}^3\text{d}^2$ -hybridized state
72. How many  $\sigma$ - and  $\pi$ - bonds are there in salicyclic acid (2-hydroxy benzoic acid)?  
 (1)  $10\sigma$ ,  $4\pi$  (2)  $16\sigma$ ,  $4\pi$  (3)  $18\sigma$ ,  $2\pi$  (4)  $16\sigma$ ,  $2\pi$
73. Which of the following pairs is (are) isostructural?  
 (1)  $\text{SF}_4$  and  $\text{SiF}_4$  (2)  $\text{SF}_6$  and  $\text{SiF}_6^{2-}$  (3)  $\text{SiF}_6^{2-}$  and  $\text{SeF}_6^{2-}$  (4)  $\text{XeO}_6^{4-}$  and  $\text{TeF}_6^{2-}$
74. The structure of  $\text{XeF}_6$  is :  
 (1) pentagonal bipyramidal (2) trigonal bipyramidal  
 (3) capped octahedral (4) square bipyramidal
75. Isostructural species are those which have the same shape and hybridisation. Among the given species identify the isostructural pairs.  
 (1)  $[\text{NF}_3]$  and  $[\text{BF}_3]$  (2)  $[\text{BF}_4^-]$  and  $[\text{NH}_4^+]$  (3)  $[\text{BCl}_3]$  and  $[\text{BrCl}_3]$  (4)  $[\text{NH}_3]$  and  $[\text{NO}_3^-]$

76. The types of hybrid orbitals of nitrogen in  $\text{NO}_2^+$ ,  $\text{NO}_3^-$  and  $\text{NH}_4^+$  respectively are expected to be  
 (1) sp,  $\text{sp}^3$  and  $\text{sp}^2$  (2) sp,  $\text{sp}^2$  and  $\text{sp}^3$  (3)  $\text{sp}^2$ , sp and  $\text{sp}^3$  (4)  $\text{sp}^2$ ,  $\text{sp}^3$  and sp
77. Which of the following angle corresponds to  $\text{sp}^2$  hybridisation?  
 (1)  $90^\circ$  (2)  $120^\circ$  (3)  $180^\circ$  (4)  $109^\circ$
78. Which of the following species has tetrahedral geometry?  
 (1)  $\text{BH}_4^-$  (2)  $\text{NH}_2^-$  (3)  $\text{CO}_3^{2-}$  (4)  $\text{H}_3\text{O}^+$
79. Carbon atoms in  $\text{C}_2(\text{CN})_4$  are :  
 (1) sp-hybridized (2)  $\text{sp}^2$ -hybridized  
 (3) sp- and  $\text{sp}^2$  hybridized (4) sp,  $\text{sp}^2$  and  $\text{sp}^3$  - hybridized
80. The shapes of  $\text{IF}_5$  and  $\text{IF}_7$  are respectively :  
 (1) square pyramidal and pentagonal bipyramidal  
 (2) octahedral and pyramidal  
 (3) trigonal bipyramidal and square antiprismatic  
 (4) distorted square planar and distorted octahedral
81.  $\text{CO}_2$  has the same geometry as :  
 (I)  $\text{HgCl}_2$  (II)  $\text{NO}_2^-$  (III)  $\text{SnCl}_4$  (IV)  $\text{C}_2\text{H}_2$   
 (1) I and III (2) II and IV (3) I and IV (4) III and IV
82. Which of the following has a geometry different from the other three species (having the same geometry)?  
 (1)  $\text{BF}_4^-$  (2)  $\text{SO}_4^{2-}$  (3)  $\text{XeF}_4$  (4)  $\text{PH}_4^+$

#### Dipole Moment :

83.  $\text{BeF}_2$  has zero dipole moment whereas  $\text{H}_2\text{O}$  has dipole moment because:  
 (1) Water is linear  
 (2)  $\text{H}_2\text{O}$  is bent  
 (3) F is more electronegative than O  
 (4) Hydrogen bonding is present in  $\text{H}_2\text{O}$
84. Which of the following molecule have zero dipole moment:  
 (1)  $\text{BF}_3$  (2)  $\text{CH}_2\text{Cl}_2$  (3)  $\text{NF}_3$  (4)  $\text{SO}_2$
85. The dipole moment of  $\text{NH}_3$  is:  
 (1) Less than dipole moment of  $\text{NCl}_3$  (2) Higher than dipole moment of  $\text{NCl}_3$   
 (3) Equal to the dipole moment of  $\text{NCl}_3$  (4) None of these
86. Which of the following species are polar:  
 (1)  $\text{C}_6\text{H}_6$  (2)  $\text{XeF}_2$  (3)  $\text{SO}_2$  (4)  $\text{SF}_4$  (5)  $\text{SF}_6$   
 Correct answer is:  
 (1) (2) & (4) (2) (1), (2) & (5) (3) (1) & (5) (4) (3) & (4)
87. Which set of molecules is polar:  
 (1)  $\text{XeF}_4$ ,  $\text{IF}_7$ ,  $\text{SO}_3$  (2)  $\text{PCl}_5$ ,  $\text{C}_6\text{H}_6$ ,  $\text{SF}_6$   
 (3)  $\text{SnCl}_2$ ,  $\text{SO}_2$ ,  $\text{NO}_2$  (4)  $\text{CO}_2$ ,  $\text{CS}_2$ ,  $\text{C}_2\text{H}_6$
88. Dipole moment is shown by:  
 (1) 1, 4 - dichlorobenzene (2) Cis 1, 2 - dichloro ethene  
 (3) Trans -1, 2 - dichloro ethene (4) benzene
89. The correct order of dipole moment is:  
 (1)  $\text{CH}_4 < \text{NF}_3 < \text{NH}_3 < \text{H}_2\text{O}$  (2)  $\text{NF}_3 < \text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$   
 (3)  $\text{NH}_3 < \text{NF}_3 < \text{CH}_4 < \text{H}_2\text{O}$  (4)  $\text{H}_2\text{O} < \text{NH}_3 < \text{NF}_3 < \text{CH}_4$

90. Which contains both polar and non-polar bonds?

- (1)  $\text{NH}_4\text{Cl}$  (2)  $\text{HCN}$  (3)  $\text{H}_2\text{O}_2$  (4)  $\text{CH}_4$

**Molecular Orbital Theory (MOT) :**

91. The number of antibonding electron pairs in  $\text{O}_2^{2-}$  molecular ion on the basis of molecular orbital theory is (at no. O = 8):

- (1) 2 (2) 3 (3) 4 (4) 5

92. Which of the following option w.r.t. increasing bond order is correct?

- (1)  $\text{NO} < \text{C}_2 < \text{O}_2^- < \text{He}_2^+$  (2)  $\text{C}_2 < \text{NO} < \text{He}_2^+ < \text{O}_2^-$   
(3)  $\text{He}_2^+ < \text{O}_2^- < \text{NO} < \text{C}_2$  (4)  $\text{He}_2^+ < \text{O}_2^- < \text{C}_2 < \text{NO}$

93. The bond-orders of the given species are such that -

- (1)  $\text{O}_2^- > \text{O}_2^+ > \text{O}_2^{2-} > \text{O}_2$  (2)  $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$   
(3)  $\text{O}_2^+ > \text{O}_2^{2-} > \text{O}_2^- > \text{O}_2$  (4)  $\text{O}_2^{2-} > \text{O}_2 > \text{O}_2^+ > \text{O}_2^-$

94. The ion that is isoelectronic with CO and having same bond order is:

- (1)  $\text{CN}^-$  (2)  $\text{O}_2^+$  (3)  $\text{O}_2^-$  (4)  $\text{N}_2^+$

95. Which of the following is paramagnetic:

- (1)  $\text{O}_2^-$  (2)  $\text{CN}^-$  (3) CO (4)  $\text{NO}^+$

96. In the following which of the two are paramagnetic:

- (1)  $\text{N}_2$  (2) CO (3)  $\text{B}_2$  (4)  $\text{NO}_2$

Correct answer is:

- (1) (1) & (3) (2) (2) & (3) (3) (3) & (4) (4) (2) & (4)

97. Increasing order of bond length in NO,  $\text{NO}^+$  and  $\text{NO}^-$  is:

- (1)  $\text{NO} > \text{NO}^- > \text{NO}^+$  (2)  $\text{NO}^+ < \text{NO} < \text{NO}^-$   
(3)  $\text{NO} < \text{NO}^+ < \text{NO}^-$  (4)  $\text{NO} < \text{NO}^+ = \text{NO}^-$

98. The paramagnetic property of oxygen is well explained by:

- (1) Molecular orbital theory (2) Resonance theory  
(3) Valence bond theory (4) VSEPR theory

99. Which of the following has fractional bond order?

- (1)  $\text{O}_2^{2+}$  (2)  $\text{O}_2^{2-}$  (3)  $\text{F}_2^{2-}$  (4)  $\text{H}_2^-$

100. Higher is the bond order, greater is:

- (1) Bond dissociation energy (2) Covalent character  
(3) Bond length (4) Paramagnetism

101. The bond order of CO molecule on the basis of molecular orbital theory is:

- (1) Zero (2) 2 (3) 3 (4) 1

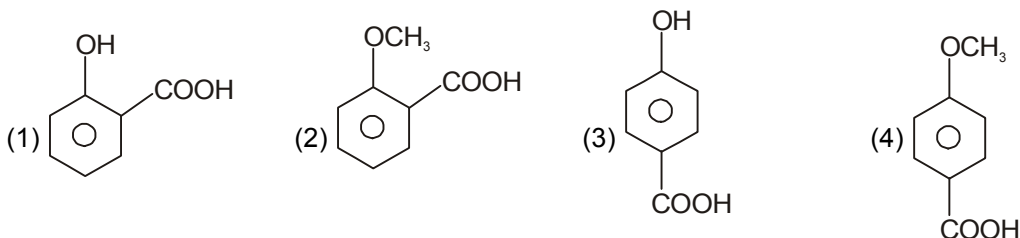
102. Which of the following group of molecules have  $2\frac{1}{2}$  bond order:

- (1)  $\text{N}_2^{2-}$ ,  $\text{O}_2^{2-}$ , CO (2)  $\text{N}_2^+$ ,  $\text{O}_2^+$ , NO (3)  $\text{C}_2^{2-}$ , BN,  $\text{O}_2$  (4)  $\text{CN}^-$ ,  $\text{NO}^+$ ,  $\text{O}_2^{+2}$

103. Which of the following species will have the minimum bond energy:
- (1)  $N_2$  (2)  $N_2^-$  (3)  $N_2^+$  (4)  $N_2^{2-}$
104. Which of the following ion has not bond order of 2.5 ?
- (1)  $O_2^-$  (2)  $O_2^+$  (3)  $N_2^+$  (4)  $N_2^-$
105.  $N_2$  and  $O_2$  are converted into monoanions,  $N_2^-$  and  $O_2^-$  respectively. Which of the following statements is wrong ?
- (1) In  $N_2^-$ , N-N bond weakens (2) In  $O_2^-$ , O-O bond order increases
- (3) In  $O_2^-$ , O-O bond order decreases (4)  $N_2^-$  becomes paramagnetic
106. Which of the following has maximum bond strength -
- (1)  $O_2$  (2)  $O_2^+$  (3)  $O_2^-$  (4)  $O_2^{2-}$

#### Hydrogen bonding :

107.  $H_2O$  boils at higher temperature than  $H_2S$ , because it is capable of forming:
- (1) Ionic bonds (2) Covalent bonds
- (3) Hydrogen bonds (4) Metallic bonds
108. Intramolecular H-bond:
- (1) Decreases Volatility (2) Increases melting point
- (3) Increases viscosity (4) Increases vapour pressure
109. Weakest hydrogen bond is:
- (1)  $O-H \cdots N$  (2)  $S-H \cdots S$  (3)  $F-H \cdots F$  (4)  $N-H \cdots N$
110. Which of the following has strongest intra molecular hydrogen bonding:



111. The high boiling point of water is due to the presence of:
- (1) Dative bond (2) Covalent bond
- (3) Hydrogen bond (4) Vander waals bond
112. Maximum no. of hydrogen bonds formed by a water molecule in ice is:
- (1) 4 (2) 3 (3) 2 (4) 1
113.  $KF$  combines with  $HF$  to form  $KHF_2$ . The compound contains the species:
- (1)  $K^+$ ,  $F^-$  and  $H^+$  (2)  $K^+$ ,  $F^-$  and  $HF$
- (3)  $K^+$  and  $[HF_2]^-$  (4)  $[KHF]^+$  and  $F_2$
114. The boiling point of a compound is raised by:
- (1) intermolecular hydrogen bonding (2) high volatility
- (3) intramolecular hydrogen bonding (4) non-polarity

115. The boiling point of p-nitrophenol is higher than that of o-nitrophenol because:  
 (1)  $\text{NO}_2$  group at p-position behaves in a different way from that at o-position  
 (2) intramolecular hydrogen bonding exists in p-nitrophenol  
 (3) there is intermolecular hydrogen bonding in p-nitrophenol  
 (4) p-nitrophenol has a higher molecular weight than o-nitrophenol
116. Which of the following compounds show intramolecular hydrogen bonding:  
 (1) o-nitrophenol      (2) p-nitrophenol      (3) phenol      (4) salicylaldehyde  
 (1) 1 & 2      (2) 1 & 3      (3) 1 & 4      (4) 2 & 3

**Integer types questions :**

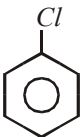
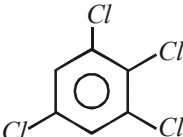
117. What is covalency of chlorine atom in second excited state ?
118. Sum of  $\sigma$  and  $\pi$  bonds in  $\text{NH}_4^+$  cation is .....
119. The molecule  $\text{ML}_x$  is planar with 6 electron pairs around M in the valence shell . The value of x is :
120. Consider the following species  $\text{NO}_3^-$ ,  $\text{SO}_4^{2-}$ ,  $\text{ClO}_3^-$ ,  $\text{SO}_3$ ,  $\text{PO}_4^{3-}$ ,  $\text{XeO}_3$ ,  $\text{CO}_3^{2-}$ ,  $\text{SO}_3^{2-}$   
 Then calculate value of  $|x - y|$ , where  
 x : Total number of species which have bond order 1.5 or greater than 1.5  
 y : Total number of species which have bond order less than 1.5
121. In  $\text{IF}_6^-$  and  $\text{TeF}_5^-$ , sum of axial d-orbitals which are used in hybridisation in both species.
122. Among the following total no. of planar species is :  
 (i)  $\text{SF}_4$       (ii)  $\text{BrF}_3$       (iii)  $\text{XeF}_2$       (iv)  $\text{IF}_5$       (v)  $\text{SbF}_4^-$       (vi)  $\text{SF}_5^-$   
 (vii)  $\text{SeF}_3^+$       (viii)  $\text{CH}_3^+$       (ix)  $\text{PCl}_4^+$
123. Calculate the value of  $X - Y$ , for  $\text{XeOF}_4$ . (X= Number of  $\sigma$  bond pair and Y = Number of lone pair on central atom)
124. Find out number of transformation among following which involves the change of hybridisation of underlined atom.  
 (a)  $\text{H}_2\underline{\text{O}} + \text{H}^+ \longrightarrow \text{H}_3\underline{\text{O}}^+$       (b)  $\text{NH}_3 + \underline{\text{B}}\text{F}_3 \longrightarrow \text{NH}_3 \cdot \underline{\text{B}}\text{F}_3$   
 (c)  $\underline{\text{Xe}}\text{F}_6 \longrightarrow \underline{\text{Xe}}\text{F}_5^+ + \text{F}^-$       (d)  $2\underline{\text{P}}\text{Cl}_5 \longrightarrow \underline{\text{P}}\text{Cl}_4^+ + \underline{\text{P}}\text{Cl}_6^-$   
 (e)  $\underline{\text{C}}\text{H}_3 - \underline{\text{C}}\text{H}_3 \longrightarrow \underline{\text{C}}\text{H}_3^- + \underline{\text{C}}\text{H}_3^+$
125. Consider the following six changes  
 (i)  $\text{NO} \longrightarrow \text{NO}^+$       (ii)  $\text{O}_2^- \longrightarrow \text{O}_2^{2-}$       (iii)  $\text{O}_2 \longrightarrow \text{O}_2^+$       (iv)  $\text{NO}^+ \longrightarrow \text{NO}^-$   
 (v)  $\text{NO}^+ \longrightarrow \text{NO}^{2+}$       (vi)  $\text{CO} \longrightarrow \text{CO}^+$   
 Then calculate value of " $c^3 - b^2 - a$ ", where a, b and c are total number of transformations in which magnetic property will be changed, bond order increases and bond order decreases respectively.
126. Total number of angle in  $\text{SeCl}_4$  which are less than  $90^\circ$ .

## EXERCISE - II

1. The compound will have lowest melting point :  
 (1) NaCl                      (2)  $\text{MgF}_2$                       (3)  $\text{AlF}_3$                       (4)  $\text{SiCl}_4$
  
2. In the electronic structure of acetic acid the number of electrons present are:  
 (1) 16 shared and 8 unshared                      (2) 8 shared and 16 unshared  
 (3) 12 shared and 12 unshared                      (4) 18 shared and 6 unshared
  
3. Amongst  $\text{NH}_3$ ,  $\text{BeCl}_2$ ,  $\text{CO}_2$  and  $\text{H}_2\text{O}$ , the non-linear molecules are :  
 (1)  $\text{BeCl}_2$  and  $\text{H}_2\text{O}$                       (2)  $\text{BeCl}_2$  and  $\text{CO}_2$   
 (3)  $\text{NH}_3$  and  $\text{H}_2\text{O}$                       (4)  $\text{NH}_3$  and  $\text{CO}_2$
  
4. Which is not correct:  
 (1) Bond angle  $\text{H-S-H} < \text{H-OH}$                       (2) Bond angle  $\text{F-O-F} < \text{Cl-O-Cl}$   
 (3) Bond angle  $\text{H-P-H} < \text{H-N-H}$                       (4) Bond angle  $\text{Cl-Sn-Cl} > \text{Cl-Hg-Cl}$
  
5. Which of the following match is not correct:  
 (1)  $\text{ICl}_2^-$  — Linear ion                      (2)  $\text{ICl}_4^-$  — Square planar ion  
 (3)  $\text{XeF}_2$  — Linear molecule                      (4)  $\text{SO}_4^{2-}$  — Trigonal planar ion
  
6. The number of unpaired electrons in paramagnetic tetrachloromagnate (II) anion is:  
 (1) 5                      (2) 2                      (3) 3                      (4) 6
  
7. The value of bond order in  $\text{NO}^+$  according to MOT is:  
 (1) 3                      (2) 2                      (3) 1                      (4) 0
  
8. Which one of the following is most polar:  
 (1)  $\text{Cl-Cl}$                       (2)  $\text{N-F}$                       (3)  $\text{C-F}$                       (4)  $\text{O-F}$
  
9. Which of the following halogens has the highest bond energy:  
 (1)  $\text{F}_2$                       (2)  $\text{Cl}_2$                       (3)  $\text{Br}_2$                       (4)  $\text{I}_2$
  
10. In the series, ethane, ethylene & acetylene, the carbon-hydrogen bond length is:  
 (1) Equal in all the three                      (2) Largest in ethane  
 (3) Smallest in ethylene                      (4) Largest in acetylene
  
11. Which of the following are isoelectronic and isostuctural:  
 $\text{NO}_3^-, \text{CO}_3^{2-}, \text{ClO}_3^-, \text{SO}_3$   
 (1)  $\text{NO}_3^-, \text{CO}_3^{2-}$                       (2)  $\text{SO}_3, \text{NO}_3^-$                       (3)  $\text{ClO}_3^-, \text{CO}_3^{2-}$                       (4)  $\text{CO}_3^{2-}, \text{ClO}_3^-$
  
12. The types of bonds present in  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  are  
 (1) electrovalent and covalent                      (2) electrovalent and coordinate covalent  
 (3) covalent and coordinate covalent                      (4) electrovalent, covalent and coordinate covalent

13. The order of increasing lattice energy of the following salt is.  
 (1)  $\text{NaCl} < \text{CaO} < \text{NaBr} < \text{BaO}$  (2)  $\text{NaBr} < \text{NaCl} < \text{BaO} < \text{CaO}$   
 (3)  $\text{NaCl} < \text{NaBr} < \text{BaO} < \text{CaO}$  (4)  $\text{NaBr} < \text{NaCl} < \text{CaO} < \text{BaO}$
14. Two elements A & B have following electronic configuration.  
 $A = 1s^2 2s^2 2p^6 3s^2$        $B = 1s^2 2s^2 2p^6 3s^2 3p^3$   
 If expected compound formed by A & B is  $A_x B_y$  then sum of x & y is  
 (1) 2 (2) 3 (3) 4 (4) 5
15. Which among the following factors is the most important in making fluorine the strongest oxidising agent ?  
 (1) Electron affinity (2) Ionisation enthalpy  
 (3) Hydration enthalpy (4) Bond dissociation energy
16. **INCORRECT** order for mentioned property is :  
 (1) Arrhenius acidic character :  $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$   
 (2) Thermal stability :  $\text{PbH}_4 > \text{SnH}_4 > \text{SiH}_4$   
 (3) Reducing character :  $\text{SbH}_3 > \text{AsH}_3 > \text{PH}_3$   
 (4) Lewis basic character :  $\text{OEt}_2 > \text{H}_2\text{O}$
17. Which of the following is most stable ?  
 (1)  $\text{Sn}^{2+}$  (2)  $\text{Ge}^{2+}$  (3)  $\text{Si}^{2+}$  (4)  $\text{Pb}^{2+}$
18. The shapes of  $\text{PCl}_4^+$ ,  $\text{PCl}_4^-$  and  $\text{AsCl}_5$  are respectively :  
 (1) square planar, tetrahedral and see-saw  
 (2) tetrahedral, see-saw and trigonal bipyramidal  
 (3) tetrahedral, square planar and pentagonal bipyramidal  
 (4) trigonal bipyramidal, tetrahedral and square pyramidal
19. An electrovalent compound does not exhibit space isomerism because of  
 (1) Presence of oppositely charged ions (2) High melting points  
 (3) Non-directional nature of the bond (4) Crystalline nature
20.  $\text{RbO}_2$  is  
 (1) Peroxide and paramagnetic (2) Peroxide and diamagnetic  
 (3) Superoxide and paramagnetic (4) Superoxide and diamagnetic
21. Bond energy of which of the following interaction is less than 8 kJ/mole.  
 (1)  $\text{Cu}^{2+}(\text{aq})$  and  $\text{NH}_3$  (2)  $\text{Xe}$  and  $\text{H}_2\text{O}$  (3)  $\text{Na}^+$  and  $[\text{BF}_4]^-$  (4)  $\text{C}_2\text{H}_5\text{OH}$  and  $\text{HF}$
22. Correct order of decreasing thermal stability is  
 (1)  $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$  (2)  $\text{PH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{SbH}_3$   
 (3)  $\text{AsH}_3 > \text{PH}_3 > \text{NH}_3 > \text{SbH}_3$  (4)  $\text{SbH}_3 > \text{AsH}_3 > \text{PH}_3 > \text{NH}_3$
23. Bond energy of a bond formed by weak dipole-dipole interaction is found to be  
 (1) Greater than 8 kJ/mole (2) Lying between 8 – 42 kJ/mole  
 (3) Less than 8 kJ/mole (4) Greater than 42 kJ/mole

24. (C—Cl) bond in  $\text{CH}_2=\text{CH}-\text{Cl}$  (vinyl chloride) is stabilised in the same way as in:  
 (1) benzyl chloride      (2) ethyl chloride      (3) chlorobenzene      (4) allyl chloride
25. Match list I with list II and select the correct answer:
- | List I (species) |                 |   |   | List II (O—N—O angle) |             |   |   |   |   |
|------------------|-----------------|---|---|-----------------------|-------------|---|---|---|---|
| A.               | $\text{NO}_2^+$ |   |   | 1.                    | $180^\circ$ |   |   |   |   |
| B.               | $\text{NO}_2$   |   |   | 2.                    | $134^\circ$ |   |   |   |   |
| C.               | $\text{NO}_2^-$ |   |   | 3.                    | $120^\circ$ |   |   |   |   |
| D.               | $\text{NO}_3^-$ |   |   | 4.                    | $115^\circ$ |   |   |   |   |
|                  |                 |   |   | 5.                    | $109^\circ$ |   |   |   |   |
|                  | A               | B | C | D                     | A           | B | C | D |   |
| (1)              | 5               | 4 | 3 | 2                     | (2)         | 5 | 2 | 4 | 3 |
| (3)              | 1               | 2 | 4 | 3                     | (4)         | 1 | 4 | 3 | 2 |
26. Which of the following fluorides of Xenon is impossible  
 (1)  $\text{XeF}_6$       (2)  $\text{XeF}_4$       (3)  $\text{XeF}_3^-$       (4)  $\text{XeF}_2$
27. The stability of +1 oxidation state increases in the sequence  
 (1)  $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$       (2)  $\text{Tl} < \text{In} < \text{Ga} < \text{Al}$   
 (3)  $\text{In} > \text{Tl} < \text{Ga} < \text{Al}$       (4)  $\text{Ga} < \text{In} < \text{Al} < \text{Tl}$
28. Hybridization of carbon atoms in  $\text{C}_3\text{O}_2$  and  $\text{C}_{12}\text{O}_9$  are respectively :  
 (1)  $\text{sp}$ ,  $\text{sp}^2$       (2)  $\text{sp}^2$ ,  $\text{sp}^3$       (3)  $\text{sp}^3$ ,  $\text{sp}$       (4)  $\text{sp}$ ,  $\text{sp}$
29. The correct order of acidic strength is  
 (1)  $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$       (2)  $\text{HCl} < \text{HBr} < \text{HF} < \text{HI}$   
 (3)  $\text{HBr} > \text{HCl} < \text{HI} < \text{HF}$       (4)  $\text{HI} < \text{HBr} < \text{HCl} < \text{HF}$
30. Select the **CORRECT** order of stability of following cations  
 (1)  $\text{Pb}^{4+} > \text{Sn}^{4+}$       (2)  $\text{Bi}^{5+} > \text{Sb}^{5+}$       (3)  $\text{In}^{3+} > \text{Tl}^{3+}$       (4)  $\text{Sn}^{2+} > \text{Sn}^{4+}$
31. Which of the following is Lewis acid-Lewis base interaction  
 (i)  $\text{B}(\text{CH}_3)_3$  and  $\text{N}(\text{CH}_3)_3$  form addition compound  
 (ii)  $\text{H}_3\text{BO}_3$  and  $\text{C}_2\text{H}_5\text{OH}$  form addition compound  
 (iii) Ammonium ion is formed by protonation of ammonia molecule  
 (iv)  $\text{BF}_3$  and  $\text{NH}_3$  form addition compound  
 (1) Only (ii), (iii) are correct      (2) Only (iv) is correct  
 (3) (ii), (iii) & (iv) are correct      (4) All are correct
32. Which element out of He, Ar, Kr, and Xe forms least numbers of compounds  
 (1) He      (2) Ar      (3) Kr      (4) Xe
33. Which of the hydrogen halides forms salts like  $\text{KHX}_2$  (where X is a halogen atom)  
 (1) HF      (2) HCl      (3) HI      (4) HBr

34. Isostructural pair will be  
 (1)  $\text{I}_3^-$  and  $\text{I}_3^+$  (2)  $\text{CO}_3^{2-}$  and  $\text{XeO}_3$  (3)  $\text{ICl}_4^-$  and  $\text{XeF}_4$  (4)  $\text{PO}_4^{3-}$  and  $\text{SeF}_4$
35. Which is the most volatile compound  
 (1) HF (2) HCl (3) HBr (4) HI
36. The tendency of  $\text{BF}_3$ ,  $\text{BCl}_3$  and  $\text{BBr}_3$  to behave as Lewis acid decreases in the sequence  
 (1)  $\text{BF}_3 > \text{BCl}_3 > \text{BBr}_3$  (2)  $\text{BCl}_3 > \text{BF}_3 > \text{BBr}_3$  (3)  $\text{BBr}_3 > \text{BCl}_3 > \text{BF}_3$  (4)  $\text{BBr}_3 > \text{BF}_3 > \text{BCl}_3$
37.  $\text{Be}(\text{OH})_2$  is insoluble in water while  $\text{Ba}(\text{OH})_2$  is highly soluble due to  
 (1) Bond order (2) Lattice energy difference  
 (3) Common ion effect (4) All
38. The dipole moment of  is 1.5 D. The dipole moment of  is :  
 (1) 0 D (2) 1.5 D (3) 2.86 D (4) 2.25 D
39. Non-polar interhalogen molecule is :  
 (1) Tetra-atomic (2) Hexa-atomic (3) Octa-atomic (4) None of these
40. The pairs of bases in DNA are held together by :  
 (1) Hydrogen bonds (2) Ionic bonds (3) Phosphate groups (4) Deoxyribose groups
41. Which of the following pair has same bond order but different magnetic behaviour  
 (1)  $\text{H}_2^+$  and  $\text{H}_2^-$  (2)  $\text{N}_2^{2+}$  and  $\text{N}_2^{2-}$  (3)  $\text{O}_2^+$  and  $\text{O}_2^-$  (4)  $\text{C}_2^+$  and  $\text{C}_2^-$
42. Increasing order of bond length in  $\text{NO}$ ,  $\text{NO}^+$  and  $\text{NO}^-$  are  
 (1)  $\text{NO} > \text{NO}^- > \text{NO}^+$  (2)  $\text{NO}^+ < \text{NO} < \text{NO}^-$  (3)  $\text{NO} < \text{NO}^+ < \text{NO}^-$  (4)  $\text{NO} < \text{NO}^+ = \text{NO}^-$
43. Bond order of  $\text{Be}_2$  is :  
 (1) 1 (2) 2 (3) 3 (4) 0
44. Peroxide bond is present in  
 (1) MgO (2) CaO (3)  $\text{Li}_2\text{O}$  (4)  $\text{BaO}_2$
45. Select pair of molecules with paramagnetic and diamagnetic nature respectively.  
 (1)  $\text{B}_2$ ,  $\text{C}_2$  (2)  $\text{C}_2$ ,  $\text{N}_2$  (3)  $\text{N}_2$ ,  $\text{O}_2$  (4)  $\text{O}_2$ ,  $\text{B}_2$
46. Correct order of melting point is :  
 (1) Diamond > Graphite >  $\text{C}_{60}$  (2) Graphite > Diamond >  $\text{C}_{60}$   
 (3)  $\text{C}_{60}$  > Graphite > Diamond (4) Diamond >  $\text{C}_{60}$  > Graphite
47. Increasing order of stability of the +2 oxidation state of ions?  
 (1)  $\text{Ca}^{2+} < \text{Ba}^{2+} < \text{Sr}^{2+}$  (2)  $\text{Pb}^{2+} < \text{Ge}^{2+} < \text{Sn}^{2+}$  (3)  $\text{Ge}^{2+} < \text{Sn}^{2+} < \text{Pb}^{2+}$  (4)  $\text{Cu}^{2+} < \text{Au}^{2+} < \text{Ag}^{2+}$
48. For  $\text{H}_2\text{O}_2$ ,  $\text{H}_2\text{S}$ ,  $\text{H}_2\text{O}$  and HF, the correct order of decreasing **extent** of hydrogen bonding is :  
 (1)  $\text{H}_2\text{O} > \text{HF} > \text{H}_2\text{O}_2 > \text{H}_2\text{S}$  (2)  $\text{H}_2\text{O} > \text{HF} > \text{H}_2\text{S} > \text{H}_2\text{O}_2$   
 (3)  $\text{HF} > \text{H}_2\text{O} > \text{H}_2\text{O}_2 > \text{H}_2\text{S}$  (4)  $\text{H}_2\text{O}_2 > \text{H}_2\text{O} > \text{HF} > \text{H}_2\text{S}$

49. Which of the following ion has a planar structure  
 (1)  $\text{SiO}_4^{4-}$  (2)  $\text{PO}_4^{3-}$  (3)  $\text{CO}_3^{2-}$  (4)  $\text{SO}_4^{2-}$
50. Which of the following statement is **incorrect** ?  
 (1) s-orbital never forms a  $\pi$ -bond.  
 (2)  $\begin{array}{c} \text{CH}_2 - \text{OH} \\ | \\ \text{CH}_2 - \text{OH} \end{array} < \begin{array}{c} \text{CH}_2 - \text{OH} \\ | \\ \text{CH} - \text{OH} \\ | \\ \text{CH}_2 - \text{OH} \end{array}$  : Order of viscosity  
 (3) A water molecule always form 2 hydrogen bonds.  
 (4) in  $\text{K}^+[\text{HF}_2]^-$  ; ionic, covalent and hydrogen bonds are present.
51. The **INCORRECT** order of boiling point is  
 (1)  $\text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te} < \text{H}_2\text{O}$  (2)  $\text{H}_2\text{S} < \text{NH}_3 < \text{HF} < \text{H}_2\text{O}$   
 (3)  $\text{PH}_3 < \text{AsH}_3 < \text{SbH}_3 < \text{NH}_3$  (4)  $\text{HCl} < \text{HBr} < \text{HI} < \text{HF}$
52. Which of the following has the highest dipole moment  
 (1)  $\text{NH}_3$  (2)  $\text{PH}_3$  (3)  $\text{SbH}_3$  (4)  $\text{AsH}_3$
53. **CORRECT** order of bond length is :  
 (1)  $\text{O}_2 > \text{O}_2^+$  (2)  $\text{N}_2 > \text{N}_2^+$  (3)  $\text{C}_2 > \text{C}_2^+$  (4)  $\text{H}_2 > \text{H}_2^+$
54. Which of the following molecule has polar as well as non polar bond  
 (1)  $\text{HCN}$  (2)  $\text{PCl}_5$  (3)  $\text{H}_2\text{O}_2$  (4) All are correct
55. If molecule  $\text{AB}_x\text{E}_y$  is planar as well as polar, then value of x and y are respectively  
 (where A = Central atom, B = surrounding atom, E = lone pair)  
 (1) 4, 2 (2) 3, 2 (3) 5, 2 (4) 3, 1
56. Which of the following halides is least stable and has doubtful existence  
 (1)  $\text{ClI}_4$  (2)  $\text{FeI}_2$  (3)  $\text{SnI}_4$  (4)  $\text{PbI}_4$
57. In which of the following arrangement, the order is not according to the property indicated against it  
 (1)  $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$  Oxidising agent  
 (2)  $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$  Basic property  
 (3)  $\text{F} > \text{Cl} > \text{Br} > \text{I}$  Electron gain enthalpy  
 (4)  $\text{C} > \text{Si} > \text{Ge} > \text{Sn}$  Ability to form  $p\pi-p\pi$  bond
58. The order of strength of hydrogen bonds is:  
 (1)  $\text{ClH}\cdots\text{Cl} > \text{NH}\cdots\text{N} > \text{OH}\cdots\text{O} > \text{FH}\cdots\text{F}$  (2)  $\text{ClH}\cdots\text{Cl} < \text{NH}\cdots\text{N} < \text{OH}\cdots\text{O} < \text{FH}\cdots\text{F}$   
 (3)  $\text{ClH}\cdots\text{Cl} < \text{NH}\cdots\text{N} > \text{OH}\cdots\text{O} > \text{FH}\cdots\text{F}$  (4)  $\text{ClH}\cdots\text{Cl} < \text{NH}\cdots\text{N} < \text{OH}\cdots\text{O} > \text{FH}\cdots\text{F}$
59. In which of the following combination, Lewis-acid base interaction does not occur :  
 (1)  $\text{H}_3\text{BO}_3 + \text{H}_2\text{O}$  (2)  $\text{CO}_2 + \text{H}_2\text{O}$  (3)  $\text{KI} + \text{I}_2$  (4)  $\text{SF}_6 + \text{H}_2\text{O}$
60. Which one of the following oxides is expected exhibit paramagnetic behaviour  
 (1)  $\text{CO}_2$  (2)  $\text{SO}_2$  (3)  $\text{ClO}_2$  (4)  $\text{SiO}_2$

## EXERCISE - III

### (Question asked in previous JEE-MAIN)

1. Arrange the following bonds according to their average bond energies in descending order :  
 $\text{C} - \text{Cl}$ ,  $\text{C} - \text{Br}$ ,  $\text{C} - \text{F}$ ,  $\text{C} - \text{I}$  [JEE-Main-Online 2020]  
 (1)  $\text{C} - \text{F} > \text{C} - \text{Cl} > \text{C} - \text{Br} > \text{C} - \text{I}$  (2)  $\text{C} - \text{I} > \text{C} - \text{Br} > \text{C} - \text{Cl} > \text{C} - \text{F}$   
 (3)  $\text{C} - \text{Br} > \text{C} - \text{I} > \text{C} - \text{Cl} > \text{C} - \text{F}$  (4)  $\text{C} - \text{Cl} > \text{C} - \text{Br} > \text{C} - \text{I} > \text{C} - \text{F}$
2. The predominant intermolecular forces present in ethyl acetate, a liquid, are : [JEE-Main-Online 2020]  
 (1) London dispersion and dipole-dipole.  
 (2) London dispersion, dipole-dipole and hydrogen bonding.  
 (3) Dipole-dipole and hydrogen bonding.  
 (4) Hydrogen bonding and London dispersion.
3. The relative strength of interionic/intermolecular forces in decreasing order is : [JEE-Main-Online 2020]  
 (1) dipole-dipole > ion-dipole > ion-ion (2) ion-dipole > ion-ion > dipole-dipole  
 (3) ion-dipole > dipole-dipole > ion-ion (4) ion-ion > ion-dipole > dipole-dipole
4. The dipole moments of  $\text{CCl}_4$ ,  $\text{CHCl}_3$  and  $\text{CH}_4$  are in the order : [JEE-Main-Online 2020]  
 (1)  $\text{CHCl}_3 < \text{CH}_4 = \text{CCl}_4$  (2)  $\text{CH}_4 = \text{CCl}_4 < \text{CHCl}_3$   
 (3)  $\text{CCl}_4 < \text{CH}_4 < \text{CHCl}_3$  (4)  $\text{CH}_4 < \text{CCl}_4 < \text{CHCl}_3$
5. The bond order and the magnetic characteristics of  $\text{CN}^-$  are : [JEE-Main-Online 2020]  
 (1)  $2\frac{1}{2}$ , paramagnetic (2)  $2\frac{1}{2}$ , diamagnetic  
 (3) 3, paramagnetic (4) 3, diamagnetic
6. Match the type of interaction in column A with the distance dependence of their interaction energy in column B : [JEE-Main-Online 2020]

| A                                | B                                |
|----------------------------------|----------------------------------|
| (I) ion-ion                      | (a) $\frac{1}{r}$                |
| (II) dipole-dipole               | (b) $\frac{1}{r^2}$              |
| (III) London dispersion          | (c) $\frac{1}{r^3}$              |
|                                  | (d) $\frac{1}{r^6}$              |
| (1) (I)-(b); (II)-(d); (III)-(c) | (2) (I)-(a); (II)-(b); (III)-(d) |
| (3) (I)-(a); (II)-(b); (III)-(c) | (4) (I)-(a); (II)-(c); (III)-(d) |
7. Among the sulphates of alkaline earth metals, the solubilities of  $\text{BeSO}_4$  and  $\text{MgSO}_4$  in water, respectively, are : [JEE-Main-Online 2020]  
 (1) poor and poor (2) High and poor (3) High and high (4) Poor and high

8. The compound that has the largest H–M–H bond angle (M = N, O, S, C), is : [JEE-Main-Online 2020]  
 (1) H<sub>2</sub>O (2) NH<sub>3</sub> (3) H<sub>2</sub>S (4) CH<sub>4</sub>
9. The shape/structure of [XeF<sub>5</sub>]<sup>–</sup> and XeO<sub>3</sub>F<sub>2</sub>, respectively, are : [JEE-Main-Online 2020]  
 (1) pentagonal planar and trigonal bipyramidal (2) octahedral and square pyramidal  
 (3) trigonal bipyramidal and pentagonal planar (4) trigonal bipyramidal and trigonal bipyramidal
10. The molecular geometry of SF<sub>6</sub> is octahedral. What is the geometry of SF<sub>4</sub> (including lone pair(s) of electrons, if any) ? [JEE-Main-Online 2020]  
 (1) Tetrahedral (2) Trigonal bipyramidal (3) Pyramidal (4) Square planar
11. In a molecule of pyrophosphoric acid, the number of P–OH, P = O and P–O–P bonds/moiety(ies) respectively are : [JEE-Main-Online 2020]  
 (1) 2, 4 and 1 (2) 3, 3 and 3 (3) 4, 2 and 0 (4) 4, 2 and 1
12. Of the species, NO, NO<sup>+</sup>, NO<sup>2+</sup> and NO<sup>–</sup>, the one with minimum bond strength is : [JEE-Main-Online 2020]  
 (1) NO<sup>+</sup> (2) NO (3) NO<sup>2+</sup> (4) NO<sup>–</sup>
13. The structure of PCl<sub>5</sub> in the solid state is : [JEE-Main-Online 2020]  
 (1) Tetrahedral [PCl<sub>4</sub>]<sup>+</sup> and octahedral [PCl<sub>6</sub>]<sup>–</sup> (2) Square planar [PCl<sub>4</sub>]<sup>+</sup> and octahedral [PCl<sub>6</sub>]<sup>–</sup>  
 (3) Square pyramidal (4) Trigonal bipyramidal
14. The reaction in which the hybridisation of the underlined atom is affected is : [JEE-Main-Online 2020]  
 (A) H<sub>3</sub>PO<sub>2</sub>  $\xrightarrow{\text{Disproportionation}}$  (B) H<sub>2</sub>SO<sub>4</sub> + NaCl  $\xrightarrow{420\text{K}}$   
 (C) NH<sub>3</sub>  $\xrightarrow{\text{H}^+}$  (D) XeF<sub>4</sub> + SbF<sub>5</sub>  $\longrightarrow$
15. The oxidation states of transition metal atoms in K<sub>2</sub>Cr<sub>2</sub>O<sub>7</sub>, KMnO<sub>4</sub> and K<sub>2</sub>FeO<sub>4</sub>, respectively, are x, y and z. The sum of x, y and z is. [JEE-Main-Online 2020]
16. The number of Cl = O bonds in perchloric acid is : [JEE-Main-Online 2020]
17. Which among the following species has unequal bond lengths? [JEE-Main-Online 2021]  
 (1) SF<sub>4</sub> (2) BF<sub>4</sub><sup>–</sup> (3) SiF<sub>4</sub> (4) XeF<sub>4</sub>
18. The correct order of bond dissociation enthalpy of halogens is [JEE-Main-Online 2021]  
 (1) I<sub>2</sub> > Br<sub>2</sub> > Cl<sub>2</sub> > F<sub>2</sub> (2) F<sub>2</sub> > Cl<sub>2</sub> > Br<sub>2</sub> > I<sub>2</sub> (3) Cl<sub>2</sub> > Br<sub>2</sub> > F<sub>2</sub> > I<sub>2</sub> (4) Cl<sub>2</sub> > F<sub>2</sub> > Br<sub>2</sub> > I<sub>2</sub>
19. According to molecular orbital theory, the species among the following that does not exist is [JEE-Main-Online 2021]  
 (1) O<sub>2</sub><sup>2–</sup> (2) He<sub>2</sub><sup>–</sup> (3) He<sub>2</sub><sup>+</sup> (4) Be<sub>2</sub>
20. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.  
**Assertion A** : Dipole-dipole interactions are the only non-covalent interactions, resulting in hydrogen bond formation.  
**Reason R** : Fluorine is the most electronegative element and hydrogen bonds in HF are symmetrical.  
 In the light of the above statements, choose the most appropriate answer from the options given below :  
 (1) Both A and R are true but R is NOT the correct explanation of A [JEE-Main-Online 2021]  
 (2) Both A and R are true and R is the correct explanation of A  
 (3) A is false but R is true  
 (4) A is true but R is false

21. Given below are two statements : [JEE-Main-Online 2021]  
**Statement-I** : o-Nitrophenol is steam volatile due to intramolecular hydrogen bonding.  
**Statement-II** : o-Nitrophenol has high melting due to hydrogen bonding.  
 In the light of the above statements, choose the most appropriate answer from the options given below :  
 (1) Statement I is false but Statement II is true  
 (2) Statement I is true but Statement II is false  
 (3) Both Statement I and Statement II is true  
 (4) Both Statement I and Statement II is false
22. The correct shape and I – I – I bond angles respectively in  $I_3^-$  ion are [JEE-Main-Online 2021]  
 (1) Distorted trigonal planar ;  $135^\circ$  and  $90^\circ$   
 (2) T-shaped ;  $180^\circ$  and  $90^\circ$   
 (3) Linear ;  $180^\circ$   
 (4) Trigonal planar ;  $120^\circ$
23. Match List-I with List-II. [JEE-Main-Online 2021]
- |     | List-I<br>(Molecule) |       | List-II<br>(Bond order) |
|-----|----------------------|-------|-------------------------|
| (a) | $Ne_2$               | (i)   | 1                       |
| (b) | $N_2$                | (ii)  | 2                       |
| (c) | $F_2$                | (iii) | 0                       |
| (d) | $O_2$                | (iv)  | 3                       |
- Choose the correct answer from the options given below :  
 (1) (a)  $\rightarrow$  (i), (b)  $\rightarrow$  (ii), (c)  $\rightarrow$  (iii), (d)  $\rightarrow$  (iv)  
 (2) (a)  $\rightarrow$  (ii), (b)  $\rightarrow$  (i), (c)  $\rightarrow$  (iv), (d)  $\rightarrow$  (iii)  
 (3) (a)  $\rightarrow$  (iv), (b)  $\rightarrow$  (iii), (c)  $\rightarrow$  (ii), (d)  $\rightarrow$  (i)  
 (4) (a)  $\rightarrow$  (iii), (b)  $\rightarrow$  (iv), (c)  $\rightarrow$  (i), (d)  $\rightarrow$  (ii)
24. In  ${}^1CH_2 = {}^2C = {}^3CH - {}^4CH_3$  molecule, the hybridization of carbon 1, 2, 3 and 4 respectively are : [JEE-Main-Online 2021]  
 (1)  $sp^2, sp^3, sp^2, sp^3$     (2)  $sp^3, sp, sp^3, sp^3$     (3)  $sp^2, sp, sp^2, sp^3$     (4)  $sp^2, sp^2, sp^2, sp^3$
25. Which of the following are isostructural pairs? [JEE-Main-Online 2021]  
 (a)  $SO_4^{2-}$  and  $CrO_4^{2-}$     (b)  $SiCl_4$  and  $TiCl_4$     (c)  $NH_3$  and  $NO_3$     (d)  $BCl_3$  and  $BrCl_3$   
 (1) c and d only    (2) a and b only    (3) a and c only    (4) b and c only
26. Out of the following, which type of interaction is responsible for the stabilisation of  $\alpha$ -helix structure of proteins? [JEE-Main-Online 2021]  
 (1) vander Waals forces    (2) Hydrogen bonding  
 (3) Ionic bonding    (4) Covalent bonding
27. Amongst the following, the linear species is : [JEE-Main-Online 2021]  
 (1)  $NO_2$     (2)  $Cl_2O$     (3)  $O_3$     (4)  $N_3^-$

28. Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)** :  
**Assertion (A)** : The H–O–H bond angle in water molecule is  $104.5^\circ$   
**Reason (R)** : The lone pair-lone pair repulsion of electrons is higher than the bond pair-bond pair repulsion.  
 In the light of the above statements, choose the correct answer from the options given below :  
**[JEE-Main-Online 2021]**
- (1) **A** is false but **R** is true  
 (2) Both **A** and **R** are true but **R** is not the correct explanation of **A**  
 (3) **A** is true but **R** is false  
 (4) Both **A** and **R** are true and **R** is the correct explanation of **A**
29. Number of paramagnetic oxides among the following given oxides is  $\text{Li}_2\text{O}$ ,  $\text{CaO}$ ,  $\text{Na}_2\text{O}_2$ ,  $\text{KO}_2$ ,  $\text{MgO}$  and  $\text{K}_2\text{O}$   
**[JEE-Main-Online 2021]**
- (1) 0 (2) 3 (3) 2 (4) 1
30. A central atom in a molecule has two lone pairs of electrons and forms three single bonds. The shape of this molecule is :  
**[JEE-Main-Online 2021]**
- (1) see- saw (2) Planar triangular (3) T-shaped (4) trigonal pyramidal
31. Which of the following compound CANNOT act as a Lewis base ?  
**[JEE-Main-Online 2021]**
- (1)  $\text{NF}_3$  (2)  $\text{PCl}_5$  (3)  $\text{SF}_4$  (4)  $\text{ClF}_3$
32. The oxidation states of nitrogen in  $\text{NO}$ ,  $\text{NO}_2$ , and  $\text{NO}_3^-$ ,  $\text{N}_2\text{O}$  are in the order of: **[JEE-Main-Online 2021]**
- (1)  $\text{NO}_3^- > \text{NO}_2 > \text{NO} > \text{N}_2\text{O}$  (2)  $\text{NO}_2 > \text{NO}_3^- > \text{NO} > \text{N}_2\text{O}$   
 (3)  $\text{N}_2\text{O} > \text{NO}_2 > \text{NO} > \text{NO}_3^-$  (4)  $\text{NO} > \text{NO}_2 > \text{N}_2\text{O} > \text{NO}_3^-$
33. A hard substance melts at high temperature and is an insulator in both solid and in molten state. This solid is most likely to be a/an : **[JEE-Main-Online 2021]**
- (1) Ionic solid (2) Molecular solid (3) Metallic solid (4) Covalent solid
34. The correct order of following 3d metal oxides, according to their oxidation numbers is :  
 (a)  $\text{CrO}_3$  (b)  $\text{Fe}_2\text{O}_3$  (c)  $\text{MnO}_2$  (d)  $\text{V}_2\text{O}_5$   
 (e)  $\text{Cu}_2\text{O}$  **[JEE-Main-Online 2021]**
- (1) (a) > (c) > (d) > (b) > (e) (2) (d) > (a) > (b) > (c) > (e)  
 (3) (a) > (d) > (c) > (b) > (e) (4) (c) > (a) > (d) > (e) > (b)
35. The water having more dissolved  $\text{O}_2$  is : **[JEE-Main-Online 2021]**
- (1) polluted water (2) water at  $80^\circ\text{C}$  (3) boiling water (4) water at  $4^\circ\text{C}$
36. The set in which compounds have different nature is : **[JEE-Main-Online 2021]**
- (1)  $\text{Be}(\text{OH})_2$  and  $\text{Al}(\text{OH})_3$  (2)  $\text{B}(\text{OH})_3$  and  $\text{Al}(\text{OH})_3$   
 (3)  $\text{B}(\text{OH})_3$  and  $\text{H}_3\text{PO}_3$  (4)  $\text{NaOH}$  and  $\text{Ca}(\text{OH})_2$
37. The hybridisations of the atomic orbitals of nitrogen in  $\text{NO}_2^-$ ,  $\text{NO}_2^+$  and  $\text{NH}_4^+$  respectively :  
**[JEE-Main-Online 2021]**
- (1)  $\text{sp}^3$ ,  $\text{sp}$  and  $\text{sp}^2$  (2)  $\text{sp}^2$ ,  $\text{sp}$  and  $\text{sp}^3$  (3)  $\text{sp}^3$ ,  $\text{sp}^2$  and  $\text{sp}$  (4)  $\text{sp}$ ,  $\text{sp}^2$  and  $\text{sp}^3$
38. In the following the correct bond order sequence is : **[JEE-Main-Online 2021]**
- (1)  $\text{O}_2 > \text{O}_2^- > \text{O}_2^{2-} > \text{O}_2^+$  (2)  $\text{O}_2^{2-} > \text{O}_2^+ > \text{O}_2^- > \text{O}_2$   
 (3)  $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$  (4)  $\text{O}_2^+ > \text{O}_2^- > \text{O}_2^{2-} > \text{O}_2$

39. Identify the species having one  $\pi$ -bond and maximum number of canonical forms from the following : [JEE-Main-Online 2021]
- (1)  $O_2$  (2)  $SO_3$  (3)  $CO_3^{2-}$  (4)  $SO_2$
40. Identify the process in which change in the oxidation state is five [JEE-Main-Online 2021]
- (1)  $C_2O_4^{2-} \rightarrow 2CO_2$  (2)  $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$  (3)  $CrO_4^{2-} \rightarrow Cr^{3+}$  (4)  $MnO_4^- + Mn^{2+}$
41. Match List - I with List - II : [JEE-Main-Online 2021]
- | List - I<br>(Species) | List - II<br>(Hybrid Orbitals) |
|-----------------------|--------------------------------|
| (a) $SF_4$            | (i) $sp^3d^2$                  |
| (b) $IF_5$            | (ii) $d^2sp^3$                 |
| (c) $NO_2^+$          | (iii) $sp^3d$                  |
| (d) $NH_4^+$          | (iv) $sp^3$                    |
|                       | (v) $sp$                       |
- Choose the correct answer from the options given below :
- (1) (a)-(iii), (b)-(i), (c)-(v) and (d)-(iv) (2) (a)-(ii), (b)-(i), (c)-(iv) and (d)-(v)  
 (3) (a)-(iv), (b)-(iii), (c)-(ii) and (d)-(v) (4) (a)-(i), (b)-(ii), (c)-(v) and (d)-(iii)
42. Which one of the following group-15 hydride is the strongest reducing agent ? [JEE-Main-Online 2021]
- (1)  $SbH_3$  (2)  $BiH_3$  (3)  $PH_3$  (4)  $AsH_3$
43. The oxidation states of 'P' in  $H_4P_2O_7$ ,  $H_4P_2O_5$  and  $H_4P_2O_6$ , respectively, are : [JEE-Main-Online 2021]
- (1) 6, 4 and 5 (2) 5, 3 and 4 (3) 7, 5 and 6 (4) 5, 4 and 3
44. Match List - I with List - II : [JEE-Main-Online 2021]
- | List - I       | List - II        |
|----------------|------------------|
| (a) NaOH       | (i) Acidic       |
| (b) $Be(OH)_2$ | (ii) Basic       |
| (c) $Ca(OH)_2$ | (iii) Amphoteric |
| (d) $B(OH)_3$  |                  |
| (e) $Al(OH)_3$ |                  |
- Choose the most appropriate answer from the option given below :
- (1) (a)-(ii), (b)-(ii), (c)-(iii), (d)-(i), (e)-(iii) (2) (a)-(ii), (b)-(iii), (c)-(ii), (d)-(i), (e)-(iii)  
 (3) (a)-(ii), (b)-(ii), (c)-(iii), (d)-(ii), (e)-(iii) (4) (a)-(ii), (b)-(i), (c)-(ii), (d)-(iii), (e)-(iii)
45. Number of Cl = O bonds in chlorous acid, chloric acid and perchloric acid respectively are : [JEE-Main-Online 2021]
- (1) 4, 1 and 0 (2) 3, 1 and 1 (3) 1, 1 and 3 (4) 1, 2 and 3
46. The bond order and magnetic behaviour of  $O_2^-$  ion are, respectively : [JEE-Main-Online 2021]
- (1) 2 and diamagnetic (2) 1.5 and paramagnetic  
 (3) 1 and paramagnetic (4) 1.5 and diamagnetic
47. The interaction energy of London forces between two particles is proportional to  $r^x$ , where r is the distance between the particles. The value of x is : [JEE-Main-Online 2021]
- (1) 3 (2) -3 (3) -6 (4) 6

48. Match List - I with List - II :

[JEE-Main-Online 2021]

**List - I**

**(Species)**

- (a)  $\text{XeF}_2$   
(b)  $\text{XeO}_2\text{F}_2$   
(c)  $\text{XeO}_3\text{F}_2$   
(d)  $\text{XeF}_4$

**List - II**

**(No. of lone pairs of electrons on the central atom)**

- (i) 0  
(ii) 1  
(iii) 2  
(iv) 3

Choose the most appropriate answer from the options given below

- (1) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)      (2) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)  
(3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)      (4) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

49. Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

**Assertion (A)** : Barium carbonate is insoluble in water and is highly stable.

**Reason (R)** : The thermal stability of the carbonates increases with increasing cationic size.

Choose the most appropriate answer from the options given below :

[JEE-Main-Online 2021]

- (1) Both **(A)** and **(R)** are true and **(R)** is the true explanation of **(A)**.  
(2) Both **(A)** and **(R)** are true but **(R)** is not the true explanation of **(A)**.  
(3) **(A)** is true but **(R)** is false.  
(4) **(A)** is false but **(R)** is true.

50. The number of S = O bonds present in sulphurous acid, peroxodisulphuric acid and pyrosulphuric acid, respectively are :

[JEE-Main-Online 2021]

- (1) 2, 3 and 4      (2) 1, 4 and 3      (3) 1, 4 and 4      (4) 2, 4 and 3

51. In polythionic acid,  $\text{H}_2\text{S}_x\text{O}_6$  ( $x = 3$  to  $5$ ) the oxidation state(s) of sulphur is/are : [JEE-Main-Online 2021]

- (1) +5 only      (2) 0 and + 5 only      (3) + 6 only      (4) + 3 and + 5 only

52. In which one of the following molecules strongest back donation of an electron pair halide to boron is expected?

[JEE-Main-Online 2021]

- (1)  $\text{BBr}_3$       (2)  $\text{BI}_3$       (3)  $\text{BCl}_3$       (4)  $\text{BF}_3$

53. The spin-only magnetic moment value of  $\text{B}_2^+$  species is \_\_\_\_\_  $\times 10^{-2}$  BM. [JEE-Main-Online 2021]

(Nearest integer) [Given :  $\sqrt{3} = 1.73$ ]

54.  $\text{AB}_3$  is an interhalogen T-shaped molecule. The number of lone pairs of electrons on A is \_\_\_\_\_. (Integer answer) [JEE-Main-Online 2021]

55. The number of species having non-pyramidal shape among the following is \_\_\_\_\_. [JEE-Main-Online 2021]

- (i)  $\text{SO}_3$       (ii)  $\text{NO}_3$       (iii)  $\text{PCl}_3$       (iv)  $\text{CO}_3^{2-}$

[JEE-Main-Online 2021]

56. According to molecular orbital theory, the number of unpaired electron(s) in  $\text{O}_2^{2-}$  is \_\_\_\_\_. [JEE-Main-Online 2021]

[JEE-Main-Online 2021]

57. The difference between bond orders of CO and  $\text{NO}^+$  is  $\frac{x}{2}$  where  $x =$  \_\_\_\_\_ (Round off to the Nearest integer) [JEE-Main-Online 2021]

[JEE-Main-Online 2021]

58. The total number of electrons in all bonding molecular orbitals of  $O_2^{2-}$  is \_\_\_\_\_.  
(Round off to the Nearest Integer). [JEE-Main-Online 2021]
59. The number of lone pairs of electrons on the central I atom in  $I_3^-$  is \_\_\_\_\_. [JEE-Main-Online 2021]
60. The number of sigma bonds in  

$$\begin{array}{c} \text{H}_3\text{C}-\text{C}=\text{CH}-\text{C}\equiv\text{C}-\text{H} \\ | \\ \text{H} \end{array}$$
is \_\_\_\_\_. [JEE-Main-Online 2021]
61. The number of species below that have two lone pairs of electrons in their central atom is. (Round off to the Nearest Integer).  
 $\text{SF}_4$ ,  $\text{BF}_4^-$ ,  $\text{ClF}_3$ ,  $\text{AsF}_3$ ,  $\text{PCl}_3$ ,  $\text{BrF}_5$ ,  $\text{XeF}_4$ ,  $\text{SF}_6$  [JEE-Main-Online 2021]
62. Identify the **incorrect** statement for  $\text{PCl}_5$  from the following. [JEE-Main-Online 2022]  
 (1) In this molecule, orbitals of phosphorous are assumed to undergo  $sp^3d$  hybridization.  
 (2) The geometry of  $\text{PCl}_5$  is trigonal bipyramidal.  
 (3)  $\text{PCl}_5$  has two axial bonds stronger than three equatorial bonds.  
 (4) The three equatorial bonds of  $\text{PCl}_5$  lie in a plane
63. The oxide which contains an odd electron at the nitrogen atom is : [JEE-Main-Online 2022]  
 (1)  $\text{N}_2\text{O}$  (2)  $\text{NO}_2$  (3)  $\text{N}_2\text{O}_3$  (4)  $\text{N}_2\text{O}_5$
64. Nitrogen gas is obtained by thermal decomposition of [JEE-Main-Online 2022]  
 (1)  $\text{Ba}(\text{NO}_3)_2$  (2)  $\text{Ba}(\text{N}_3)_2$  (3)  $\text{NaNO}_2$  (4)  $\text{NaNO}_3$
65. Consider the ions/molecules [JEE-Main-Online 2022]  
 $\text{O}_2^+$ ,  $\text{O}_2$ ,  $\text{O}_2^-$ ,  $\text{O}_2^{2-}$   
 For increasing bond order the **correct** option is :  
 (1)  $\text{O}_2^{2-} < \text{O}_2^- < \text{O}_2 < \text{O}_2^+$  (2)  $\text{O}_2^- < \text{O}_2^{2-} < \text{O}_2 < \text{O}_2^+$   
 (3)  $\text{O}_2^- < \text{O}_2^{2-} < \text{O}_2^+ < \text{O}_2$  (4)  $\text{O}_2^- < \text{O}_2^+ < \text{O}_2^{2-} < \text{O}_2$
66. Choose the **correct** stability order of group 13 elements in their +1 oxidation state. [JEE-Main-Online 2022]  
 (1)  $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$  (2)  $\text{Tl} < \text{In} < \text{Ga} < \text{Al}$  (3)  $\text{Al} < \text{Ga} < \text{Tl} < \text{In}$  (4)  $\text{Al} < \text{Tl} < \text{Ga} < \text{In}$
67. Bonding in which of the following diatomic molecule(s) become(s) stronger, on the basis of MO Theory, by removal of an electron?  
 (A)  $\text{NO}$  (B)  $\text{N}_2$  (C)  $\text{O}_2$  (D)  $\text{C}_2$  (E)  $\text{B}_2$   
 Choose the **most appropriate** answer from the options given below : [JEE-Main-Online 2022]  
 (1) (A), (B), (C) only (2) (B), (C), (E) only (3) (A), (C) only (4) (D) only
68. Number of electron deficient molecules among the following  
 $\text{PH}_3$ ,  $\text{B}_2\text{H}_6$ ,  $\text{CCl}_4$ ,  $\text{NH}_3$ ,  $\text{LiH}$  and  $\text{BCl}_3$  is [JEE-Main-Online 2022]  
 (1) 0 (2) 1 (3) 2 (4) 3

69. The **correct** order of bond orders of  $C_2^{2-}$ ,  $N_2^{2-}$ ,  $O_2^{2-}$  is, respectively : [JEE-Main-Online 2022]  
 (1)  $C_2^{2-} < N_2^{2-} < O_2^{2-}$  (2)  $O_2^{2-} < N_2^{2-} < C_2^{2-}$   
 (3)  $C_2^{2-} < O_2^{2-} < N_2^{2-}$  (4)  $N_2^{2-} < C_2^{2-} < O_2^{2-}$
70. Arrange the following in the decreasing order of their covalent character : [JEE-Main-Online 2022]  
 (a) LiCl (b) NaCl (c) KCl (d) CsCl  
 Choose the most appropriate answer from the options given below :  
 (1) (a) > (c) > (b) > (d) (2) (b) > (a) > (c) > (d)  
 (3) (a) > (b) > (c) > (d) (4) (a) > (b) > (d) > (c)
71. Number of lone pair(s) of electrons on central atom and the shape of  $BrF_3$  molecule respectively, are [JEE-Main-Online 2022]  
 (1) 0, triangular planar (2) 1, pyramidal (3) 2, bent T-shape (4) 1, bent T-shape
72.  $PCl_5$  is well known, but  $NCl_5$  is not. Because, [JEE-Main-Online 2022]  
 (1) Nitrogen is less reactive than phosphorous  
 (2) Nitrogen doesn't have d-orbitals in its valence shell  
 (3) Catenation tendency is weaker in nitrogen than phosphorous  
 (4) Size of phosphorous is larger than nitrogen
73. The **correct** order of increasing intermolecular hydrogen bond strength is [JEE-Main-Online 2022]  
 (1)  $HCN < H_2O < NH_3$  (2)  $HCN < CH_4 < NH_3$   
 (3)  $CH_4 < HCN < NH_3$  (4)  $CH_4 < NH_3 < HCN$
74. The number of bridged oxygen atoms present in compound B formed from the following reactions is [JEE-Main-Online 2022]  

$$Pb(NO_3)_2 \xrightarrow{673K} A + PbO + O_2$$

$$A \xrightarrow{\text{Dimerise}} B$$
  
 (1) 0 (2) 1 (3) 2 (4) 3
75. Consider the species  $CH_4$ ,  $NH_4^+$  and  $BH_4^-$ . Choose the **correct** option with respect to these species. [JEE-Main-Online 2022]  
 (1) They are isoelectronic and only two have tetrahedral structures  
 (2) They are isoelectronic and all have tetrahedral structures  
 (3) Only two are isoelectronic and all have tetrahedral structures  
 (4) Only two are isoelectronic and only two have tetrahedral structures
76. Given below are the oxides :  $Na_2O$ ,  $As_2O_3$ ,  $N_2O$ ,  $NO$  and  $Cl_2O_7$   
 Number of amphoteric oxides is : [JEE-Main-Online 2022]  
 (1) 0 (2) 1 (3) 2 (4) 3
77. The **correct** order of melting points of hydrides of group 16 elements is : [JEE-Main-Online 2022]  
 (1)  $H_2S < H_2Se < H_2Te < H_2O$  (2)  $H_2O < H_2S < H_2Se < H_2Te$   
 (3)  $H_2S < H_2Te < H_2Se < H_2O$  (4)  $H_2Se < H_2S < H_2Te < H_2O$

78. Identify the **correct** statement for  $B_2H_6$  from those given below : [JEE-Main-Online 2022]  
 (A) In  $B_2H_6$ , all B-H bonds are equivalent. (B) In  $B_2H_6$ , there are four 3-centre-2-electron bonds.  
 (C)  $B_2H_6$  is a Lewis acid. (D)  $B_2H_6$  can be synthesized from both  $BF_3$  and  $NaBH_4$ .  
 (E)  $B_2H_6$  is a planar molecule.  
 Choose the most appropriate answer from the options given below : [JEE-Main-Online 2022]  
 (1) (A) and (E) only (2) (B), (C) and (E) only  
 (3) (C) and (D) only (4) (C) and (E) only
79. The most stable trihalide of nitrogen is : [JEE-Main-Online 2022]  
 (1)  $NF_3$  (2)  $NCl_3$  (3)  $NBr_3$  (4)  $NI_3$
80. Among the following, basic oxide is : [JEE-Main-Online 2022]  
 (1)  $SO_3$  (2)  $SiO_2$  (3)  $CaO$  (4)  $Al_2O_3$
81. Among the given oxides of nitrogen ;  $N_2O$ ,  $N_2O_3$ ,  $N_2O_4$  and  $N_2O_5$ , the number of compound/(s) having N-N bond is : [JEE-Main-Online 2022]  
 (1) 1 (2) 2 (3) 3 (4) 4
82. Which of the following oxoacids of sulphur contains "S" in two different oxidation states? [JEE-Main-Online 2022]  
 (1)  $H_2S_2O_3$  (2)  $H_2S_2O_6$  (3)  $H_2S_2O_7$  (4)  $H_2S_2O_8$
83. In the structure of  $SF_4$ , the lone pair of electrons on S is in. [JEE-Main-Online 2022]  
 (1) Equatorial position and there are two lone pair-bond pair repulsions at  $90^\circ$   
 (2) Equatorial position and there are three lone pair-bond pair repulsions at  $90^\circ$   
 (3) Axial position and there are three lone pair-bond pair repulsion at  $90^\circ$   
 (4) Axial position and there are two lone pair-bond pair repulsion at  $90^\circ$
84. Given below are two statements : [JEE-Main-Online 2022]  
**Statement I** : The pentavalent oxide of group-15 element,  $E_2O_5$ , is less acidic than trivalent oxide,  $E_2O_3$ , of the same element.  
**Statement II** : The acidic character of trivalent oxide of group 15 elements,  $E_2O_3$ , decreases down the group.  
 In light of the above statements, choose **most appropriate** answer from the options given below :  
 (1) Both Statement I and Statement II are true  
 (2) Both Statement I and Statement II are false  
 (3) Statement I true, but Statement II is false  
 (4) Statement I false, but Statement II is true
85. Based upon VSEPR theory, match the shape (geometry) of the molecules in **List-I** with the molecules in **List-II** and select the most **appropriate** option. [JEE-Main-Online 2022]
- | List-I (Shape)                                     | List-II (Molecules)                                |
|----------------------------------------------------|----------------------------------------------------|
| (A) T-shaped                                       | (I) $XeF_4$                                        |
| (B) Trigonal planar                                | (II) $SF_4$                                        |
| (C) Square planar                                  | (III) $ClF_3$                                      |
| (D) See-saw                                        | (IV) $BF_3$                                        |
| (1) (A) – (I), (B) – (II), (C) – (III), (D) – (IV) | (2) (A) – (III), (B) – (IV), (C) – (I), (D) – (II) |
| (3) (A) – (III), (B) – (IV), (C) – (II), (D) – (I) | (4) (A) – (IV), (B) – (III), (C) – (I), (D) – (II) |

86. Match List-I with List-II :

[JEE-Main-Online 2022]

| List-I (Oxide)              | List-II (Nature) |
|-----------------------------|------------------|
| (A) $\text{Cl}_2\text{O}_7$ | (I) Amphoteric   |
| (B) $\text{Na}_2\text{O}$   | (II) Basic       |
| (C) $\text{Al}_2\text{O}_3$ | (III) Neutral    |
| (D) $\text{N}_2\text{O}$    | (IV) Acidic      |

Choose the **correct** answer from the options given below :

- |                            |                            |
|----------------------------|----------------------------|
| (1) A-IV, B-III, C-I, D-II | (2) A-IV, B-II, C-I, D-III |
| (3) A-II, B-IV, C-III, D-I | (4) A-I, B-II, C-III, D-IV |

87. Match List-I with List-II :

| List-I<br>(molecule) | List-II<br>(hybridization; shape)                   |
|----------------------|-----------------------------------------------------|
| A. $\text{XeO}_3$    | I. $\text{sp}^3\text{d}$ ; linear                   |
| B. $\text{XeF}_2$    | II. $\text{sp}^3$ ; pyramidal                       |
| C. $\text{XeOF}_4$   | III. $\text{sp}^3\text{d}^3$ ; distorted octahedral |
| D. $\text{XeF}_6$    | IV. $\text{sp}^3\text{d}^2$ ;square pyramidal       |

Choose the correct answer from the options given below :

[JEE-Main-Online 2022]

- |                            |                            |
|----------------------------|----------------------------|
| (1) A-II, B-I, C-IV, D-III | (2) A-II, B-IV, C-III, D-I |
| (3) A-IV, B-II, C-III, D-I | (4) A-IV, B-II, C-I, D-III |

88. Match List-I with List-II.

| List-I<br>(Compound)      | List-II<br>(Shape)         |
|---------------------------|----------------------------|
| (A) $\text{BrF}_5$        | (I) bent                   |
| (B) $[\text{CrF}_6]^{3-}$ | (II) square pyramidal      |
| (C) $\text{O}_3$          | (III) trigonal bipyramidal |
| (D) $\text{PCl}_5$        | (IV) octahedral            |

Choose the correct answer from the options given below :

[JEE-Main-Online 2022]

- |                                                    |                                                    |
|----------------------------------------------------|----------------------------------------------------|
| (1) (A) – (I), (B) - (II), (C) - (III), (D) - (IV) | (2) (A) - (IV), (B) - (III), (C) - (II), (D) - (I) |
| (3) (A) - (II), (B) - (IV), (C) - (I), (D) - (III) | (4) (A) - (III), (B) - (IV), (C) - (II), (D) - (I) |

89. Match List-I with List-II

[JEE-Main-Online 2022]

| List-I                                                     | List-II                                    |
|------------------------------------------------------------|--------------------------------------------|
| (A) $\Psi_{\text{MO}} = \Psi_{\text{A}} - \Psi_{\text{B}}$ | (I) Dipole moment                          |
| (B) $\mu = Q \times r$                                     | (II) Bonding molecular orbital             |
| (C) $\frac{N_{\text{b}} - N_{\text{a}}}{2}$                | (III) Anti-bonding molecular orbital       |
| (D) $\Psi_{\text{MO}} = \Psi_{\text{A}} + \Psi_{\text{B}}$ | (IV) Bond order                            |
| (1) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)                 | (2) (A)-(III), (B)-(IV), (C)-(I), (D)-(II) |
| (3) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)                 | (4) (A)-(III), (B)-(IV), (C)-(II), (D)-(I) |

90. Given below are two statements :
- Statement-I** : The chlorides of Be and Al have Cl-bridged structure. Both are soluble in organic solvents and act as Lewis bases.
- Statement-II** : Hydroxides of Be and Al dissolve in excess alkali to give beryllate and aluminate ions.
- In the light of the above statements. Choose the correct answer from the options given below.
- [JEE-Main-Online 2022]**
- (1) Both statement I and Statement II are true  
 (2) Both statement I and Statement II are false  
 (3) Statement I is true but Statement II is false  
 (4) Statement I is false but Statement II is true
91. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**
- Assertion A** : Zero orbital overlap is an out of phase overlap.
- Reason** : It results due to different orientation/direction of approach of orbitals.
- In the light of the above statements. Choose the correct answer from the options given below
- [JEE-Main-Online 2022]**
- (1) Both A and R are true and R is the correct explanation of A  
 (2) Both A and R are true but R is NOT the correct explanation of A  
 (3) A is true but R is false  
 (4) A is false but R is true
92. Given below are two statements.
- Statement-I** : Stannane is an example of a molecular hydride.
- Statement-II** : Stannane is a planar molecule.
- In the light of the above statement, choose the most appropriate answer from the options given below :
- [JEE-Main-Online 2022]**
- (1) Both Statement I and Statement II are true.      (2) Both Statement I and Statement II are false.  
 (3) Statement I is true but Statement II is false.      (4) Statement I is false but Statement II is true.
93. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.
- Assertion A** : Boric acid is a weak acid
- Reason R** : Boric acid is not able to release  $H^+$  ion on its own. It receives  $OH^-$  ion from water and releases  $H^+$  ion.
- In the light of the above statements, choose the most appropriate answer from the options given below.
- [JEE-Main-Online 2022]**
- (1) Both A and R are correct and R is the correct explanation of A  
 (2) Both A and R are correct but R is NOT the correct explanation of A  
 (3) A is correct but R is not correct  
 (4) A is not correct but R is correct
94. Given below are two statements : one is labelled as **Assertion (1)** and the other is labelled as **Reason (R)**
- Assertion (A)** : Boron is unable to form  $BF_6^{3-}$
- Reason (R)** : Size of B is very small.
- In the light of the above statements, choose the correct answer from the options given below :
- [JEE-Main-Online 2022]**
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)  
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)  
 (3) (A) is true but (R) is false  
 (4) (A) is false but (R) is true

95. Given below are two statements.  
**Statement-I** :  $O_2$ ,  $Cu^{2+}$  and  $Fe^{3+}$  are weakly attracted by magnetic field and are magnetized in the same direction as magnetic field.  
**Statement-II** :  $NaCl$  and  $H_2O$  are weakly magnetized in opposite direction to magnetic field.  
 In the light of the above statements, choose the most appropriate answer form the options given below :  
**[JEE-Main-Online 2022]**
- (1) Both Statement I and Statement II are correct.  
 (2) Both Statement I and Statement II are incorrect.  
 (3) Statement I is correct but Statement II is incorrect.  
 (4) Statement I is incorrect but Statement II is correct.
96. The total number of Mn = O bonds in  $Mn_2O_7$  is \_\_\_\_\_.  
**[JEE-Main-Online 2022]**
- (1) 4 (2) 5 (3) 6 (4) 3
97. Number of lone pairs of electrons in the central atom of  $SCl_2$ ,  $O_3$ ,  $ClF_3$  and  $SF_6$ , respectively, are :  
**[JEE-Main-Online 2022]**
- (1) 0, 1, 2 and 2 (2) 2, 1, 2 and 0 (3) 1, 2, 2 and 0 (4) 2, 1, 2 and 0
98. Which oxoacid of phosphorous has the highest number of oxygen atoms present in its chemical formula?  
**[JEE-Main-Online 2022]**
- (1) Pyrophosphorous acid (2) Hypophosphoric acid  
 (3) Phosphoric acid (4) Pyrophosphoric acid
99. Arrange the following in increasing order of their covalent character.  
 (A)  $CaF_2$  (B)  $CaCl_2$  (C)  $CaBr_2$  (D)  $CaI_2$   
 Choose the correct answer from the options given below.  
**[JEE-Main-Online 2022]**
- (1)  $B < A < C < D$  (2)  $A < B < C < D$   
 (3)  $A < B < D < C$  (4)  $A < C < B < D$
100. The total number of acidic oxides from the following list is :  $NO$ ,  $N_2O$ ,  $B_2O_3$ ,  $N_2O_5$ ,  $CO$ ,  $SO_3$ ,  $P_4O_{10}$   
**[JEE-Main-Online 2022]**
- (1) 3 (2) 4 (3) 5 (4) 6
101. Which of the following pair of molecules contain odd electron molecule and an expanded octet molecule?  
**[JEE-Main-Online 2022]**
- (1)  $BCl_3$  and  $SF_6$  (2)  $NO$  and  $H_2SO_4$  (3)  $SF_6$  and  $H_2SO_4$  (4)  $BCl_3$  and  $NO$
102. The sum of number of lone pairs of electrons present on the central atoms of  $XeO_3$ ,  $XeOF_4$  and  $XeF_6$  is \_\_\_\_\_.  
**[JEE-Main-Online 2022]**
103. According to MO theory, number of species/ions from the following having identical bond order is \_\_\_\_ :  
 $CN^-$ ,  $NO^+$ ,  $O_2$ ,  $O_2^+$ ,  $O_2^{2+}$   
**[JEE-Main-Online 2022]**
104. Among the following species  
 $N_2$ ,  $N_2^+$ ,  $N_2^-$ ,  $N_2^{2-}$ ,  $O_2$ ,  $O_2^+$ ,  $O_2^-$ ,  $O_2^{2-}$   
 the number of species showing diamagnetism is  
**[JEE-Main-Online 2022]**
105. Consider,  $PF_5$ ,  $BrF_5$ ,  $PCl_3$ ,  $SF_6$ ,  $[ICl_4]^-$ ,  $ClF_3$  and  $IF_5$ .  
 Amongst the above molecule(s)/ion(s), the number of molecule(s)/ion(s) having  $sp^3d^2$  hybridisation is \_\_\_\_\_.  
**[JEE-Main-Online 2022]**

106. Consider the following sulphur based oxoacids.  
 $\text{H}_2\text{SO}_3$ ,  $\text{H}_2\text{SO}_4$ ,  $\text{H}_2\text{S}_2\text{O}_8$  and  $\text{H}_2\text{S}_2\text{O}_7$ .  
 Amongst these oxoacids, the number of those with peroxo(O-O) bond is \_\_\_\_\_.  
**[JEE-Main-Online 2022]**
107. Amongst the following the number of oxide(s) which are paramagnetic in nature is  
 $\text{Na}_2\text{O}$ ,  $\text{KO}_2$ ,  $\text{NO}_2$ ,  $\text{N}_2\text{O}$ ,  $\text{ClO}_2$ ,  $\text{NO}$ ,  $\text{SO}_2$ ,  $\text{Cl}_2\text{O}$  **[JEE-Main-Online 2022]**
108. The number of paramagnetic species among the following is \_\_\_\_\_.  
 $\text{B}_2$ ,  $\text{Li}_2$ ,  $\text{C}_2$ ,  $\text{C}_2^-$ ,  $\text{O}_2^{2-}$ ,  $\text{O}_2^+$  and  $\text{He}_2^+$  **[JEE-Main-Online 2022]**
109. The number of interhalogens from the following having square pyramidal structure is :  
 $\text{ClF}_3$ ,  $\text{IF}_7$ ,  $\text{BrF}_5$ ,  $\text{BrF}_3$ ,  $\text{I}_2\text{Cl}_6$ ,  $\text{IF}_5$ ,  $\text{ClF}$ ,  $\text{ClF}_5$  **[JEE-Main-Online 2022]**
110. The number of molecule(s) or ion(s) from the following having non-planar structure is \_\_\_\_\_.  
 $\text{NO}_3^-$ ,  $\text{H}_2\text{O}_2$ ,  $\text{BF}_3$ ,  $\text{PCl}_3$ ,  $\text{XeF}_4$ ,  $\text{SF}_4$ ,  $\text{XeO}_3$ ,  $\text{PH}_4^+$ ,  $\text{SO}_3$ ,  $[\text{Al}(\text{OH})_4]^-$  **[JEE-Main-Online 2022]**
111. Amongst  $\text{BeF}_2$ ,  $\text{BF}_3$ ,  $\text{H}_2\text{O}$ ,  $\text{NH}_3$ ,  $\text{CCl}_4$  and  $\text{HCl}$ , the number of molecules with non-zero net dipole moment is \_\_\_\_\_.  
**[JEE-Main-Online 2022]**
112. The hybridization of P exhibited in  $\text{PF}_5$  is  $\text{sp}_x\text{d}_y$ . The value of y is \_\_\_\_\_.  
**[JEE-Main-Online 2022]**
113. The difference in oxidation state of chromium in chromate and dichromate salts is \_\_\_\_\_.  
**[JEE-Main-Online 2022]**
114. Amongst  $\text{SF}_4$ ,  $\text{XeF}_4$ ,  $\text{CF}_4$  and  $\text{H}_2\text{O}$ , the number of species with two lone pair of electrons is \_\_\_\_\_.  
**[JEE-Main-Online 2022]**
115. Resonance in carbonate ion ( $\text{CO}_3^{2-}$ ) is **[JEE-Main-Online 2023]**
- 
- Which of the following is true?
- (1) It is possible to identify each structure individually by some physical or chemical method.
  - (2) All these structures are in dynamic equilibrium with each other.
  - (3)  $\text{CO}_3^{2-}$  has a single structure i.e., resonance hybrid of the above three structures.
  - (4) Each structure exists for equal amount of time.
116. The Lewis acid character of boron tri halides follows the order ; **[JEE-Main-Online 2023]**
- |                                                               |                                                               |
|---------------------------------------------------------------|---------------------------------------------------------------|
| (1) $\text{BF}_3 > \text{BCl}_3 > \text{BBr}_3 > \text{BO}_3$ | (2) $\text{BCl}_3 > \text{BF}_3 > \text{BBr}_3 > \text{BI}_3$ |
| (3) $\text{BI}_3 > \text{BBr}_3 > \text{BCl}_3 > \text{BF}_3$ | (4) $\text{BBr}_3 > \text{BI}_3 > \text{BCl}_3 > \text{BF}_3$ |

117. Match List-I with List-II

**List-I**

- (A)  $\text{XeF}_4$   
(B)  $\text{SF}_4$   
(C)  $\text{NH}_4^+$   
(D)  $\text{BrF}_3$

**List-II**

- (I) See-saw  
(II) Square planar  
(III) Bent T-shaped  
(IV) Tetrahedral

Choose the correct answer from the options given below :

[JEE-Main-Online 2023]

- (1) A-II B-I C-III D-IV  
(2) A-IV B-I C-II D-III  
(3) A-II B-I C-IV D-III  
(4) A-IV B-III C-II D-I

118. Boric acid is solid, whereas  $\text{BF}_3$  is gas at room temperature because of [JEE-Main-Online 2023]

- (1) Strong van der Waal's interaction in Boric acid (2) Strong hydrogen bond in Boric acid  
(3) Strong ionic bond in Boric acid (4) Strong covalent bond in  $\text{BF}_3$

119. Bond dissociation energy of "E-H" bond of the " $\text{H}_2\text{E}$ " hydrides of group 16 elements (given below), follows order.

- (A) O (B) S (C) Se (D) Te

Choose the correct from the options given below :

[JEE-Main-Online 2023]

- (1)  $A > B > C > D$  (2)  $D > C > B > A$   
(3)  $A > B > D > C$  (4)  $B > A > C > D$

120. The correct order of hydration enthalpies is :

- (A)  $\text{K}^+$  (B)  $\text{Rb}^+$  (C)  $\text{Mg}^{2+}$  (D)  $\text{Cs}^+$  (E)  $\text{Ca}^{2+}$

Choose the correct answer from the options given below :

[JEE-Main-Online 2023]

- (1)  $C > E > A > D > B$  (2)  $C > A > E > B > D$   
(3)  $C > E > A > B > D$  (4)  $E > C > A > B > D$

121. Match List-I with List-II

**List-I**

(molecules/ions)

- (A)  $\text{IF}_7$   
(B)  $\text{ICl}_4^-$   
(C)  $\text{XeF}_6$   
(D)  $\text{XeF}_2$

**List-II**

(No. of lone pairs of  $e^-$  on central atom)

- (I) Three  
(II) One  
(III) Two  
(IV) Zero

Choose the correct answer from the options given below :

[JEE-Main-Online 2023]

- (1) A-IV B-I C-II D-III  
(2) A-II B-III C-IV D-I  
(3) A-IV B-III C-II D-I  
(4) A-II B-I C-IV D-III

122. For  $\text{OF}_2$  molecule consider the following :

- (A) Number of lone pairs on oxygen is 2. (B) FOF angle is less than  $104.5^\circ$ .  
(C) Molecule is bent 'V' shaped. (D) Molecular geometry is linear.

Correct options are :

[JEE-Main-Online 2023]

- (1) A, B, C only (2) A, B, D only (3) C, D, E only (4) A, C, D only

123. The bond dissociation energy is highest for

[JEE-Main-Online 2023]

- (1)  $\text{Br}_2$  (2)  $\text{F}_2$  (3)  $\text{I}_2$  (4)  $\text{Cl}_2$

124. What is the number of unpaired electron (s) in the highest occupied molecular orbital of the following species :  $N_2$  ;  $N_2^+$  ;  $O_2$  ;  $O_2^+$  ? [JEE-Main-Online 2023]  
 (1) 0, 1, 0, 1 (2) 2, 1, 2, 1 (3) 0, 1, 2, 1 (4) 2, 1, 0, 1
125. The correct order of bond enthalpy ( $\text{kJ mol}^{-1}$ ) is : [JEE-Main-Online 2023]  
 (1)  $C - C > Si - Si > Ge - Ge > Sn - Sn$  (2)  $C - C > Si - Si > Sn - Sn > Ge - Ge$   
 (3)  $Si - Si > C - C > Sn - Sn > Ge - Ge$  (4)  $Si - Si > C - C > Ge - Ge > Sn - Sn$
126. The magnetic behavior of  $Li_2O$ ,  $Na_2O_2$  and  $KO_2$ , respectively, are [JEE-Main-Online 2023]  
 (1) paramagnetic, diamagnetic and paramagnetic  
 (2) diamagnetic, diamagnetic and paramagnetic  
 (3) diamagnetic, paramagnetic and diamagnetic  
 (4) paramagnetic, paramagnetic and diamagnetic
127. O – O bond length in  $H_2O_2$  is  $\underline{X}$  than the O – O bond length in  $F_2O_2$ . The O – H bond length in  $H_2O_2$  is  $\underline{Y}$  than that of the O – F bond in  $F_2O_2$ .  
 Choose the correct option for  $\underline{X}$  and  $\underline{Y}$  from those given below : [JEE-Main-Online 2023]  
 (1) X - longer, Y - longer (2) X - longer, Y - shorter  
 (3) X - shorter, Y - shorter (4) X - shorter, Y - longer
128. Order of Covalent bond :  
 (A)  $KF > KI$  ;  $LiF > KF$  (B)  $KF < KI$  ;  $LiF > KF$   
 (C)  $SnCl_4 > SnCl_2$  ;  $CuCl > NaCl$  (D)  $LiF > KF$  ;  $CuCl < NaCl$   
 (E)  $KF < KI$  ;  $CuCl > NaCl$   
 Choose the correct answer from the options given below : [JEE-Main-Online 2023]  
 (1) B, C only (2) C, E only (3) B, C, E only (4) A, B only
129. Which of the Phosphorus oxoacid can create silver mirror from  $AgNO_3$  solution? [JEE-Main-Online 2023]  
 (1)  $(HPO_3)_n$  (2)  $H_4P_2O_7$  (3)  $H_4P_2O_6$  (4)  $H_4P_2O_5$
130. **Statement I** : Dipole moment is a vector quantity and by convention it is depicted by a small arrow with tail on the negative centre and head pointing towards the positive centre.  
**Statement II** : The crossed arrow of the dipole moment symbolizes the direction of the shift of charges in the molecules.  
 In the light of the above statements, choose the most appropriate answer from the options given below. [JEE-Main-Online 2023]  
 (1) Both Statement-I and Statement-II are correct  
 (2) Both Statement-I and Statement-II are incorrect  
 (3) Statement-I is correct but Statement-II is incorrect  
 (4) Statement-I is incorrect but Statement-II is correct
131. According to MO theory the bond orders for  $O_2^{2-}$ , CO and  $NO^+$  respectively, are [JEE-Main-Online 2023]  
 (1) 1, 2 and 3 (2) 1, 3 and 2 (3) 1, 3 and 3 (4) 2, 3 and 3
132. The alkaline earth metal sulphate(s) which are readily soluble in water is/are :  
 (A)  $BeSO_4$  (B)  $MgSO_4$  (C)  $CaSO_4$  (D)  $SrSO_4$   
 (E)  $BaSO_4$   
 Choose the **correct** answer from the options given below : [JEE-Main-Online 2023]  
 (1) B only (2) B and C (3) A and B (4) A only

133. For compound having the formula  $\text{Ga}[\text{AlCl}_4]$ , the correct option from the following is  
[JEE-Main-Online 2023]
- Cl forms bond with both Al and Ga in  $\text{Ga}[\text{AlCl}_4]$
  - Oxidation state of Ga in the salt  $\text{GaAlCl}_4$  is +3.
  - Ga is more electronegative than Al and is present as a cationic part of the salt  $\text{GaAlCl}_4$
  - Ga is coordinated with Cl in  $\text{GaAlCl}_4$
134. Match List-I with List-II :
- | List-I (Oxide)             | List-II (Type of bond) |
|----------------------------|------------------------|
| (A) $\text{N}_2\text{O}_4$ | (I) 1 N = O bond       |
| (B) $\text{NO}_2$          | (II) 1 N–O–N bond      |
| (C) $\text{N}_2\text{O}_5$ | (III) 1 N–N bond       |
| (D) $\text{N}_2\text{O}$   | (IV) 1 N=N / N=N bond  |
- Choose the correct answer from the options given below : [JEE-Main-Online 2023]
- A – III, B – I, C – II, D – IV
  - A – III, B – I, C – IV, D – II
  - A – II, B – I, C – III, D – IV
  - A – II, B – IV, C – III, D – I
135. Match List-I with List-II :
- | List-I (Species)          | List-II (Geometry / Shape) |
|---------------------------|----------------------------|
| A. $\text{H}_3\text{O}^+$ | I. Tetrahedral             |
| B. Acetylide anion        | II. Linear                 |
| C. $\text{NH}_4^+$        | III. Pyramidal             |
| D. $\text{ClO}_2^-$       | IV. Bent                   |
- Choose the correct answer from the options given below : [JEE-Main-Online 2023]
- A-III, B-IV, C-II, D-I
  - A-III, B-IV, C-I, D-II
  - A-III, B-II, C-I, D-IV
  - A-III, B-I, C-II, D-IV
136. Given below are two statements :
- Statement-I** :  $\text{SO}_2$  and  $\text{H}_2\text{O}$  both possess V-shaped structure.
- Statement-II** : The bond angle of  $\text{SO}_2$  is less than that of  $\text{H}_2\text{O}$ .
- In the light of the above statements, choose the most appropriate answer from the options given below:
- [JEE-Main-Online 2023]
- Statement I is incorrect but Statement II is correct
  - Both Statement I and Statement II are correct
  - Statement I is correct but Statement II is incorrect
  - Both Statement I and Statement II are incorrect
137. The covalency and oxidation state of boron in  $[\text{BF}_4]^-$ , are  
[JEE-Main-Online 2023]
- 3 and 5
  - 4 and 3
  - 3 and 4
  - 4 and 4
138. In which of the following processes, the bond order increases and paramagnetic character changes to diamagnetic one?  
[JEE-Main-Online 2023]
- $\text{O}_2 \rightarrow \text{O}_2^+$
  - $\text{N}_2 \rightarrow \text{N}_2^+$
  - $\text{O}_2 \rightarrow \text{O}_2^{2-}$
  - $\text{NO} \rightarrow \text{NO}^+$
139. The number of P – O – P bonds in  $\text{H}_4\text{P}_2\text{O}_7$ ,  $(\text{HPO}_3)_3$  and  $\text{P}_4\text{O}_{10}$  are respectively [JEE-Main-Online 2023]
- 1, 3, 6
  - 0, 3, 4
  - 0, 3, 6
  - 1, 2, 4

140. Consider the following statements:  
 (A)  $\text{NF}_3$  molecule has a trigonal planar structure.  
 (B) Bond length of  $\text{N}_2$  is shorter than  $\text{O}_2$ .  
 (C) Isoelectronic molecules or ions have identical bond order.  
 (D) Dipole moment of  $\text{H}_2\text{S}$  is higher than that of water molecule.  
 Choose the correct answer from the options given below: [JEE-Main-Online 2023]  
 (1) (C) and (D) are correct (2) (B) and (C) are correct  
 (3) (A) and (B) are correct (4) (A) and (D) are correct
141. Group-13 elements react with  $\text{O}_2$  in amorphous form to form oxides of type  $\text{M}_2\text{O}_3$  (M = element). Which among the following is the most basic oxide? [JEE-Main-Online 2023]  
 (1)  $\text{B}_2\text{O}_3$  (2)  $\text{Al}_2\text{O}_3$  (3)  $\text{Tl}_2\text{O}_3$  (4)  $\text{Ga}_2\text{O}_3$
142. The bond order and magnetic property of acetylide ion are same as that of [JEE-Main-Online 2023]  
 (1)  $\text{O}_2^-$  (2)  $\text{O}_2^+$  (3)  $\text{N}_2^+$  (4)  $\text{NO}^+$
143. Given below are two statements :  
**Statement-I** :  $\text{SbCl}_3$  is more covalent than  $\text{SbCl}_5$   
**Statement-II** : The higher oxides of halogens also tend to be more stable than the lower ones.  
 In the light of the above statements, choose the most appropriate answer from the options given below [JEE-Main-Online 2023]  
 (1) Both Statement I and Statement II are incorrect  
 (2) Both statement I and Statement II are correct  
 (3) Statement I is incorrect but Statement II is correct  
 (4) Statement I is correct but Statement II is incorrect
144. Structures of  $\text{BeCl}_2$  in solid state, vapour phase and at very high temperature respectively are : [JEE-Main-Online 2023]  
 (1) Polymeric, Dimeric, Monomeric (2) Dimeric, Polymeric, Monomeric  
 (3) Polymeric, Monomeric, Dimeric (4) Monomeric, Dimeric, Polymeric
145. Which one of the following pairs is an example of polar molecular solids? [JEE-Main-Online 2023]  
 (1)  $\text{HCl}$  (s),  $\text{AlN}$  (s) (2)  $\text{MgO}$  (s),  $\text{SO}_2$  (s)  
 (3)  $\text{SO}_2$  (s),  $\text{NH}_3$  (s) (4)  $\text{SO}_2$  (s),  $\text{CO}_2$  (s)
146. Arrange the following gases in increasing order of van der Waals constant 'a'  
 A. Ar B.  $\text{CH}_4$  C.  $\text{H}_2\text{O}$  D.  $\text{C}_6\text{H}_6$   
 Choose the correct option from the following. [JEE-Main-Online 2023]  
 (1) B, C, D and A (2) D, C, B and A  
 (3) A, B, C and D (4) C, D, B and A
147. The ratio of sigma and  $\pi$  bonds present in pyrophosphoric acid is \_\_\_\_\_. [JEE-Main-Online 2023]
148. The total number of isoelectronic species from the given set is \_\_\_\_\_.  
 $\text{O}^{2-}$ ,  $\text{F}^-$ ,  $\text{Al}$ ,  $\text{Mg}^{2+}$ ,  $\text{Na}^+$ ,  $\text{O}^+$ ,  $\text{Mg}$ ,  $\text{Al}^{3+}$ ,  $\text{F}$  [JEE-Main-Online 2023]
149. The maximum number of lone pairs of electron on the central atom from the following species is \_\_\_\_\_.  
 $\text{ClO}_3^-$ ,  $\text{XeF}_4$ ,  $\text{SF}_4$  and  $\text{I}_3^-$  [JEE-Main-Online 2023]
150. If the formula of Borax is  $\text{Na}_2\text{B}_4\text{O}_x(\text{OH})_y \cdot z\text{H}_2\text{O}$ , then  $x + y + z =$  \_\_\_\_\_. [JEE-Main-Online 2023]
151. The number of bent-shaped molecule/s from the following is \_\_\_\_\_.  
 $\text{N}_3^-$ ,  $\text{NO}_2^-$ ,  $\text{I}_3^-$ ,  $\text{O}_3$ ,  $\text{SO}_2$  [JEE-Main-Online 2023]

152. The number of following factors which affect the percent covalent character of the ionic bond is \_\_\_\_\_.  
 (i) Polarising power of cation (ii) Extent of distortion of anion  
 (iii) Polarisability of the anion (iv) Polarising power of anion  
**[JEE-Main-Online 2023]**
153. The number of species having a square planar shape from the following is \_\_\_\_\_.  
 $\text{XeF}_4$ ,  $\text{SF}_4$ ,  $\text{SiF}_4$ ,  $\text{BF}_4^-$ ,  $\text{BrF}_4^-$ ,  $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ,  $[\text{FeCl}_4]^{2-}$ ,  $[\text{PtCl}_4]^{2-}$   
**[JEE-Main-Online 2023]**
154. The sum of lone pairs present on the central atom of the interhalogen  $\text{IF}_5$  and  $\text{IF}_7$  is \_\_\_\_\_.  
**[JEE-Main-Online 2023]**
155. The number of species from the following which have square pyramidal structure is \_\_\_\_\_.  
 $\text{PF}_5$ ,  $\text{BrF}_4^-$ ,  $\text{IF}_5$ ,  $\text{BrF}_5$ ,  $\text{XeOF}_4$ ,  $\text{ICl}_4^-$   
**[JEE-Main-Online 2023]**
156. The number of species from the following carrying a single lone pair on central atom Xenon is \_\_\_\_\_.  
 $\text{XeF}_5^+$ ,  $\text{XeO}_3$ ,  $\text{XeO}_2\text{F}_2$ ,  $\text{XeF}_5^-$ ,  $\text{XeO}_3\text{F}_2$ ,  $\text{XeOF}_4$ ,  $\text{XeF}_4$   
**[JEE-Main-Online 2023]**
157. In the following reactions, the total number of oxygen atoms in X and Y is \_\_\_\_\_.  
 $\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{X}$   
 $\text{Cl}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2\text{Y}$   
**[JEE-Main-Online 2023]**
158. The number of molecules from the following which contain only two lone pair of electrons is \_\_\_\_\_.  
 $\text{H}_2\text{O}$ ,  $\text{N}_2$ ,  $\text{CO}$ ,  $\text{XeF}_4$ ,  $\text{NH}_3$ ,  $\text{NO}$ ,  $\text{CO}_2$ ,  $\text{F}_2$   
**[JEE-Main-Online 2023]**
159. Sum of oxidation states of bromine in bromic acid and perbromic acid is \_\_\_\_\_.  
**[JEE-Main-Online 2023]**
160. Sum of  $\pi$ -bonds present in peroxodisulphuric acid and pyrosulphuric acid is \_\_\_\_\_.  
**[JEE-Main-Online 2023]**
161. The number of molecules or ions from the following, which do not have odd number of electrons are \_\_\_\_\_.  
 (A)  $\text{NO}_2$  (B)  $\text{ICl}_4^-$  (C)  $\text{BrF}_3$  (D)  $\text{ClO}_2$   
 (E)  $\text{NO}_2^+$  (F)  $\text{NO}$   
**[JEE-Main-Online 2023]**
162. The total number of lone pairs of electrons on oxygen atoms of ozone is \_\_\_\_\_.  
**[JEE-Main-Online 2023]**
163. On heating,  $\text{LiNO}_3$  gives how many compounds among the following? \_\_\_\_\_.  
 $\text{Li}_2\text{O}$ ,  $\text{N}_2$ ,  $\text{O}_2$ ,  $\text{LiNO}_2$ ,  $\text{NO}_2$   
**[JEE-Main-Online 2023]**
164. The number of given orbitals which have electron density along the axis is \_\_\_\_\_.  
 $p_x, p_y, p_z, d_{xy}, d_{yz}, d_{xz}, d_{z^2}, d_{x^2-y^2}$   
**[JEE-Main-Online 2023]**
165. Amongst the following, the number of species having the linear shape is \_\_\_\_\_.  
 $\text{XeF}_2$ ,  $\text{I}_3^+$ ,  $\text{C}_3\text{O}_2$ ,  $\text{I}_3^-$ ,  $\text{CO}_2$ ,  $\text{SO}_2$ ,  $\text{BeCl}_2$  and  $\text{BCl}_2^+$   
**[JEE-Main-Online 2023]**
166. Total number of acidic oxides among  
 $\text{N}_2\text{O}_3$ ,  $\text{NO}_2$ ,  $\text{N}_2\text{O}$ ,  $\text{Cl}_2\text{O}_7$ ,  $\text{SO}_2$ ,  $\text{CO}$ ,  $\text{CaO}$ ,  $\text{Na}_2\text{O}$  and  $\text{NO}$  is \_\_\_\_\_. **[JEE-Main-Online 2023]**
167. Element not showing variable oxidation state is : **[JEE-Main-Online 2024]**  
 (1) Chlorine (2) Fluorine (3) Bromine (4) Iodine
168. Consider the oxides of group 14 elements  
 $\text{SiO}_2$ ,  $\text{GeO}_2$ ,  $\text{SnO}_2$ ,  $\text{PbO}_2$ ,  $\text{CO}$  and  $\text{GeO}$ . The amphoteric oxides are **[JEE-Main-Online 2024]**  
 (1)  $\text{SnO}_2$ ,  $\text{PbO}_2$  (2)  $\text{GeO}$ ,  $\text{GeO}_2$  (3)  $\text{SiO}_2$ ,  $\text{GeO}_2$  (4)  $\text{SnO}_2$ ,  $\text{CO}$

169. Which of the following is least ionic ? [JEE-Main-Online 2024]  
 (1)  $\text{BaCl}_2$  (2)  $\text{AgCl}$  (3)  $\text{KCl}$  (4)  $\text{CoCl}_2$
170. The linear combination of atomic orbitals to form molecular orbitals takes place only when the combining atomic orbitals  
 (a) have the same energy  
 (b) have the minimum overlap  
 (c) have same symmetry about the molecular axis  
 (d) have different symmetry about the molecular axis  
 Choose the **most appropriate** from the options given below : [JEE-Main-Online 2024]  
 (1) b and d only (2) a and c only (3) a, b, c only (4) b, c, d only
171. The difference in energy between the actual structure and the lowest energy resonance structure for the given compound is : [JEE-Main-Online 2024]  
 (1) Ionization energy (2) Electromeric energy  
 (3) Hyperconjugation energy (4) Resonance energy
172. Arrange the bonds in order of increasing ionic character in the molecules.  $\text{LiF}$ ,  $\text{K}_2\text{O}$ ,  $\text{N}_2$ ,  $\text{SO}_2$  and  $\text{ClF}_3$  : [JEE-Main-Online 2024]  
 (1)  $\text{LiF} < \text{K}_2\text{O} < \text{ClF}_3 < \text{SO}_2 < \text{N}_2$  (2)  $\text{ClF}_3 < \text{N}_2 < \text{SO}_2 < \text{K}_2\text{O} < \text{LiF}$   
 (3)  $\text{N}_2 < \text{SO}_2 < \text{ClF}_3 < \text{K}_2\text{O} < \text{LiF}$  (4)  $\text{N}_2 < \text{ClF}_3 < \text{SO}_2 < \text{K}_2\text{O} < \text{LiF}$
173. Choose the correct statements about the hydrides of group 15 elements.  
 (I) The stability of the hydrides decreases in the order  $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{BiH}_3$   
 (II) The reducing ability of the hydride increases in the order  $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3 < \text{BiH}_3$   
 (III) Among the hydrides,  $\text{NH}_3$  is strong reducing agent while  $\text{BiH}_3$  is mild reducing agent  
 (IV) The basicity of the hydrides increases in the order  $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3 < \text{BiH}_3$   
 Choose the most appropriate from the options given below : [JEE-Main-Online 2024]  
 (1) I and IV only (2) I and II only (3) II and III only (4) III and IV only
174. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**:  
**Assertion A** :  $\text{H}_2\text{Te}$  is more acidic than  $\text{H}_2\text{S}$ .  
**Reason R** : Bond dissociation enthalpy of  $\text{H}_2\text{Te}$  is lower than  $\text{H}_2\text{S}$ .  
 In the light of the above statements, choose the **most appropriate** from the options given below : [JEE-Main-Online 2024]  
 (1) A is false but R is true  
 (2) Both A and R are true but R is NOT the correct explanation of A.  
 (3) A is true but R is false  
 (4) Both A and R are true and R is the correct explanation of A
175. Given below are two statements :  
**Statement - I** : Since Fluorine is more electronegative than nitrogen, the net dipole moment of  $\text{NF}_3$  is greater than  $\text{NH}_3$ .  
**Statement - II** : In  $\text{NH}_3$ , the orbital dipole due to lone pair and the dipole moment of  $\text{NH}$  bonds are in opposite direction, but in  $\text{NF}_3$  the orbital dipole due to lone pair and dipole moments of  $\text{N} - \text{F}$  bonds are in same direction.  
 In the light of the above statements, choose the **most appropriate** from the options given below : [JEE-Main-Online 2024]  
 (1) Both Statement I and Statement - II are false (2) Both Statement I and Statement - II are true  
 (3) Statement I is false but Statement - II is true (4) Statement I is true but Statement - II is false

176. Given below are two statements :

**Statement I :** Noble gases have very high boiling points

**Statement II :** Noble gases are monoatomic gases. They are held together by strong dispersion forces. Because of this they are liquefied at very low temperature. Hence, they have very high boiling points.

In the light of the above statements, choose the **correct answer** from the options given below :

[JEE-Main-Online 2024]

- (1) Both Statement I and Statement II are false
- (2) Both Statement I and Statement II are true
- (3) Statement I is true but Statement II is false
- (4) Statement I is false but Statement II is true

177. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A) :**  $\text{PH}_3$  has lower boiling point than  $\text{NH}_3$ .

**Reason (R) :** In liquid state  $\text{NH}_3$  molecules are associated through vander Waal's forces, but  $\text{PH}_3$  molecules are associated through hydrogen bonding.

In the light of the above statements, choose the most appropriate answer from the options given below :

[JEE-Main-Online 2024]

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is not correct but (R) is correct
- (3) Both (A) and (R) are correct but (R) is the correct explanation of (A)
- (4) (A) is correct but (R) is not correct

178. Given below are two statements :

**Statement (I) :** A  $\pi$  bonding MO has lower electron density above and below the inter-nuclear axis.

**Statement (II) :** The  $\pi^*$  antibonding MO has a node between the nuclei.

In the light of the above statements, choose the most appropriate answer from the options given below :

[JEE-Main-Online 2024]

- (1) Both Statement I and Statement II are true
- (2) Statement I is true but Statement II is false
- (3) Statement I is false but Statement II is true
- (4) Both Statement I and Statement II are false

179. Match List - I with List - II

**List - I**

**Molecule**

- (1)  $\text{BrF}_5$
- (2)  $\text{H}_2\text{O}$
- (3)  $\text{ClF}_3$
- (4)  $\text{SF}_4$

**List - II**

**Shape**

- (I) T-shape
- (II) See saw
- (III) Bent
- (IV) Square pyramidal

Choose the correct answer from the options given below :

[JEE-Main-Online 2024]

- (1) (1) – (IV) , (2) - (III) , (3) - (I) , (4) - (II)
- (2) (1) – (II) , (2) - (I) , (3) - (III) , (4) - (IV)
- (3) (1) – (III) , (2) - (IV) , (3) - (I) , (4) - (II)
- (4) (1) – (I) , (2) - (II) , (3) - (IV) , (4) - (III)

180. The total number of molecules with zero dipole moment among  $\text{CH}_4$  ,  $\text{BF}_3$  ,  $\text{H}_2\text{O}$  ,  $\text{HF}$  ,  $\text{NH}_3$  ,  $\text{CO}_2$  and  $\text{SO}_2$  is\_\_\_\_\_.

[JEE-Main-Online 2024]

181. The number of species from the following in which the central atom uses  $sp^3$  hybrid orbitals in its bonding is \_\_\_\_\_.  
 $NH_3$ ,  $SO_2$ ,  $SiO_2$ ,  $BeCl_2$ ,  $CO_2$ ,  $H_2O$ ,  $CH_4$ ,  $BF_3$  [JEE-Main-Online 2024]
182. The total number of anti bonding molecular orbitals, formed from 2s and 2p atomic orbitals in a diatomic molecule is \_\_\_\_\_. [JEE-Main-Online 2024]
183. The total number of molecular orbitals formed from 2s and 2p atomic orbitals of a diatomic molecule is \_\_\_\_\_. [JEE-Main-Online 2024]
184. The number of species from the following which are paramagnetic and with bond order equal to one is \_\_\_\_\_.  
 $H_2$ ,  $He_2^+$ ,  $O_2^+$ ,  $N_2^{2-}$ ,  $O_2^{2-}$ ,  $F_2$ ,  $Ne_2^+$ ,  $B_2$  [JEE-Main-Online 2024]
185. Number of compounds with one lone pair of electrons on central atom amongst following is \_\_\_\_\_.  
 $O_3$ ,  $H_2O$ ,  $SF_4$ ,  $ClF_3$ ,  $NH_3$ ,  $BrF_5$ ,  $XeF_4$  [JEE-Main-Online 2024]
186. Sum of bond order of CO and  $NO^+$  is \_\_\_\_\_. [JEE-Main-Online 2024]
187. From the given list, the number of compounds with +4 oxidation state of Sulphur is \_\_\_\_\_.  
 $SO_3$ ,  $H_2SO_3$ ,  $SOCl_2$ ,  $SF_4$ ,  $BaSO_4$ ,  $H_2S_2O_7$  [JEE-Main-Online 2024]
188. The number of non-polar molecules from the following is \_\_\_\_\_. [JEE-Main-Online 2024]  
 $HF$ ,  $H_2O$ ,  $SO_2$ ,  $H_2$ ,  $CO_2$ ,  $CH_4$ ,  $NH_3$ ,  $HCl$ ,  $CHCl_3$ ,  $BF_3$
189. The lowest oxidation number of an atom in a compound  $A_2B$  is -2. The number of electrons in its valence shell is \_\_\_\_\_. [JEE-Main-Online 2024]
190. A diatomic molecule has a dipole moment of 1.2 D. If the bond distance is  $1\text{\AA}$ , then fractional charge on each atom is \_\_\_\_\_  $\times 10^{-1}$  esu.  
 (Given :  $1\text{ D} = 10^{-18}$  esucm) [JEE-Main-Online 2024]
191. The shape of carbocation is : [JEE-Main-Online 2024]  
 (1) tetrahedral (2) trigonal planar (3) diagonal (4) diagonal pyramidal
192. The correct increasing order for bond angles among  $BF_3$ ,  $PF_3$  and  $ClF_3$  is: [JEE-Main-Online 2024]  
 (1)  $PF_3 < BF_3 < ClF_3$  (2)  $BF_3 < PF_3 < ClF_3$   
 (3)  $ClF_3 < PF_3 < BF_3$  (4)  $BF_3 = PF_3 < ClF_3$
193. The **incorrect** statement regarding ethyne is : [JEE-Main-Online 2024]  
 (1) Ethyne is linear  
 (2) The C – C bonds in ethyne is shorter than that in ethene  
 (3) The carbon – carbon bonds in ethyne is weaker than that in ethene  
 (4) Both carbons are  $sp$  hybridised
194. When  $\Psi_A$  and  $\Psi_B$  are the wave functions of atomic orbitals, then  $\sigma^*$  is represented by : [JEE-Main-Online 2024]  
 (1)  $\Psi_A + 2\Psi_B$  (2)  $\Psi_A - 2\Psi_B$  (3)  $\Psi_A - \Psi_B$  (4)  $\Psi_A + \Psi_B$
195. The number of species from the following that have pyramidal geometry around the central atom is \_\_\_\_\_. [JEE-Main-Online 2024]  
 $S_2O_3^{2-}$ ,  $SO_4^{2-}$ ,  $SO_3^{2-}$ ,  $S_2O_7^{2-}$   
 (1) 1 (2) 3 (3) 2 (4) 4

196. The correct statement/s about Hydrogen bonding is/are  
 (A) Hydrogen bonding exists when H is covalently bonded to the highly electronegative atom  
 (B) Intermolecular H bonding is present in o-nitro phenol  
 (C) Intramolecular H bonding is present in HF  
 (D) The magnitude of H bonding depends on the physical state of the compound  
 (E) H-bonding has powerful effect on the structure and properties of compounds  
 Choose the correct answer from the options given below : **[JEE-Main-Online 2024]**  
 (1) A only (2) A, D, E only (3) A, B, D only (4) A, B, C only
197. Which one of the following molecules has maximum dipole moment? **[JEE-Main-Online 2024]**  
 (1)  $\text{CH}_4$  (2)  $\text{NF}_3$  (3)  $\text{NH}_3$  (4)  $\text{PF}_5$
198. Number of molecules/ions from the following in which the central atom is involved in  $\text{sp}^3$  hybridization is \_\_\_\_\_.  
 $\text{NO}_3^-$ ,  $\text{BCl}_3$ ,  $\text{ClO}_2^-$ ,  $\text{ClO}_3$  **[JEE-Main-Online 2024]**  
 (1) 3 (2) 4 (3) 2 (4) 1
199. In which one of the following pairs the central atoms exhibit  $\text{sp}^2$  hybridization ? **[JEE-Main-Online 2024]**  
 (1)  $\text{BF}_3$  and  $\text{NO}_2^-$  (2)  $\text{NH}_2^-$  and  $\text{H}_2\text{O}$  (3)  $\text{NH}_2^-$  and  $\text{BF}_3$  (4)  $\text{H}_2\text{O}$  and  $\text{NO}_2$
200. The statement(s) that are correct about the species  $\text{O}^{2-}$ ,  $\text{F}^-$ ,  $\text{Na}^+$  and  $\text{Mg}^{2+}$ .  
 (A) All are isoelectronic (B) All have the same nuclear charge  
 (C)  $\text{O}^{2-}$  has the largest ionic radii (D)  $\text{Mg}^{2+}$  has the smallest ionic radii  
 Choose the most appropriate answer from the options given below : **[JEE-Main-Online 2024]**  
 (1) (A), (C) and (D) only (2) (B), (C) and (D) only  
 (3) (A), (B), (C) and (D) (4) (C) and (D) only
201. Number of  $\sigma$  and  $\pi$  bonds present in ethylene molecule is respectively : **[JEE-Main-Online 2024]**  
 (1) 5 and 2 (2) 4 and 1 (3) 3 and 1 (4) 5 and 1
202. Given below are two statements:  
**Statement (I)** : The oxidation state of an element in a particular compound is the charge acquired by its atom on the basis of electron gain enthalpy consideration from other atoms in the molecule.  
**Statement (II)** :  $p\pi - p\pi$  bond formation is more prevalent to second period elements over other periods.  
 In the light of the above statements, choose the most appropriate answer from the options given below : **[JEE-Main-Online 2024]**  
 (1) Both **Statement I** and **Statement II** are incorrect  
 (2) Both **Statement I** and **Statement II** are correct  
 (3) **Statement I** is correct but **Statement II** is incorrect  
 (4) **Statement I** is incorrect but **Statement II** is correct

203. Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.  
**Assertion (A)** :  $\text{NH}_3$  and  $\text{NF}_3$  molecule have pyramidal shape with a lone pair of electrons on nitrogen atom. The resultant dipole moment of  $\text{NH}_3$  is greater than that of  $\text{NF}_3$ .  
**Reason (R)** : In  $\text{NH}_3$ , the orbital dipole due to lone pair is in the same direction as the resultant dipole moment of the N – H bonds. F is the most electronegative element.

In the light of the above statements, choose the correct answer from the options given below :

[JEE-Main-Online 2024]

- (1) (A) is false but (R) is true  
 (2) (A) is true but (R) is false  
 (3) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)  
 (4) Both (A) and (R) are true and (R) is the correct explanation of (A)

204. Match List-I with List-II

**List-I (Molecule)**

**List-II (Shape)**

- |                    |                           |
|--------------------|---------------------------|
| (A) $\text{NH}_3$  | (I) Square pyramid        |
| (B) $\text{BrF}_5$ | (II) Tetrahedral          |
| (C) $\text{PCl}_5$ | (III) Trigonal pyramidal  |
| (D) $\text{CH}_4$  | (IV) Trigonal bipyramidal |

Choose the correct answer from the options given below :

[JEE-Main-Online 2024]

- |                                    |                                    |
|------------------------------------|------------------------------------|
| (1) A - III, B - IV, C - I, D - II | (2) A - IV, B - III, C - I, D - II |
| (3) A - II, B - IV, C - I, D - III | (4) A - III, B - I, C - IV, D - II |

205. Match List-I and List-II

**List-I  
(Hybridization)**

**List-II  
(Orientation in Space)**

- |                             |                          |
|-----------------------------|--------------------------|
| (A) $\text{sp}^3$           | (I) Trigonal bipyramidal |
| (B) $\text{dsp}^2$          | (II) Octahedral          |
| (C) $\text{sp}^3\text{d}$   | (III) Tetrahedral        |
| (D) $\text{sp}^3\text{d}^2$ | (IV) Square planar       |

Choose the correct answer from the options given below :

[JEE-Main-Online 2024]

- |                                    |                                    |
|------------------------------------|------------------------------------|
| (1) A - II, B - I, C - IV, D - III | (2) A - IV, B - III, C - I, D - II |
| (3) A - III, B - I, C - IV, D - II | (4) A - III, B - IV, C - I, D - II |

206. Match List-I and List-II

**List-I  
(Molecule / Species)**

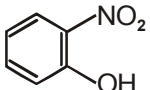
**List-II  
(Property / Shape)**

- |                              |                   |
|------------------------------|-------------------|
| (A) $\text{SO}_2\text{Cl}_2$ | (I) Paramagnetic  |
| (B) NO                       | (II) Diamagnetic  |
| (C) $\text{NO}_2^-$          | (III) Tetrahedral |
| (D) $\text{I}_3^-$           | (IV) Linear       |

Choose the correct answer from the options given below :

[JEE-Main-Online 2024]

- |                                    |                                    |
|------------------------------------|------------------------------------|
| (1) A - III, B - I, C - II, D - IV | (2) A - III, B - IV, C - II, D - I |
| (3) A - IV, B - I, C - III, D - II | (4) A - II, B - III, C - I, D - IV |

- 207. List-I (Compound/Species) List-II (Shape/Geometry)**
- (A)  $\text{SF}_4$  (I) Tetrahedral  
 (B)  $\text{BrF}_3$  (II) Pyramidal  
 (C)  $\text{BrO}_3^-$  (III) See saw  
 (D)  $\text{NH}_4^+$  (IV) Bent T-Shape
- Choose the correct answer from the options given below : **[JEE-Main-Online 2024]**
- (1) A - III, B - IV, C - II, D - I (2) A - II, B - III, C - I, D - IV  
 (3) A - III, B - II, C - IV, D - I (4) A - II, B - IV, C - III, D - I
- 208. List-I List-II**
- (A)  $\text{ICl}$  (I) T-shape  
 (B)  $\text{ICl}_3$  (II) Square pyramidal  
 (C)  $\text{ClF}_5$  (III) Pentagonal bipyramidal  
 (D)  $\text{IF}_7$  (IV) Linear
- Choose the **correct** answer from the options given below : **[JEE-Main-Online 2024]**
- (1) (A) - (I), (B) - (III), (C) - (II), (D) - (IV) (2) (A) - (IV), (B) - (I), (C) - (II), (D) - (III)  
 (3) (A) - (IV), (B) - (III), (C) - (II), (D) - (I) (4) (A) - (I), (B) - (IV), (C) - (III), (D) - (II)
- 209.** Number of molecules / species from the following having one unpaired electron is \_\_\_\_\_.  
 $\text{O}_2$ ,  $\text{O}_2^{-1}$ ,  $\text{NO}$ ,  $\text{CN}^{-1}$ ,  $\text{O}_2^{2-}$  **[JEE-Main-Online 2024]**
- 210.** The total number of species from the following in which one unpaired electron is present, is \_\_\_\_\_.  
 $\text{N}_2$ ,  $\text{O}_2$ ,  $\text{C}_2^-$ ,  $\text{O}_2^-$ ,  $\text{H}_2^+$ ,  $\text{CN}^-$ ,  $\text{He}_2^+$  **[JEE-Main-Online 2024]**
- 211.** Total number of electrons present in ( $\pi$ ) molecular orbitals of  $\text{O}_2$ ,  $\text{O}_2^+$  and  $\text{O}_2^-$  is \_\_\_\_\_.  
**[JEE-Main-Online 2024]**
- 212.** Number of molecules having bond order 2 from the following molecules is \_\_\_\_\_.  
 $\text{C}_2$ ,  $\text{O}_2$ ,  $\text{Be}_2$ ,  $\text{Li}_2$ ,  $\text{Ne}_2$ ,  $\text{N}_2$ ,  $\text{He}_2$  **[JEE-Main-Online 2024]**
- 213.** Number of compounds / species from the following with non-zero dipole moment is \_\_\_\_\_.  
 $\text{BeCl}_2$ ,  $\text{BCl}_3$ ,  $\text{NF}_3$ ,  $\text{XeF}_4$ ,  $\text{CCl}_4$ ,  $\text{H}_2\text{O}$ ,  $\text{H}_2\text{S}$ ,  $\text{HBr}$ ,  $\text{CO}_2$ ,  $\text{H}_2$ ,  $\text{HCl}$  **[JEE-Main-Online 2024]**
- 214.** In the lewis dot structure for  $\text{NO}_2^-$ , total number of valence electrons around nitrogen is \_\_\_\_\_.  
**[JEE-Main-Online 2024]**
- 215.** Number of compounds from the following with zero dipole moment is \_\_\_\_\_.  
 $\text{HF}$ ,  $\text{H}_2$ ,  $\text{H}_2\text{S}$ ,  $\text{CO}_2$ ,  $\text{NH}_3$ ,  $\text{BF}_3$ ,  $\text{CH}_4$ ,  $\text{CHCl}_3$ ,  $\text{SiF}_4$ ,  $\text{H}_2\text{O}$ ,  $\text{BeF}_2$  **[JEE-Main-Online 2024]**
- 216.** Number of molecules from the following which can exhibit hydrogen bonding is \_\_\_\_\_. (nearest integer)  
 $\text{CH}_3\text{OH}$ ,  $\text{H}_2\text{O}$ ,  $\text{C}_2\text{H}_6$ ,  $\text{C}_6\text{H}_6$ , ,  $\text{HF}$ ,  $\text{NH}_3$  **[JEE-Main-Online 2024]**
- 217.** Total number of species from the following with central atom utilising  $\text{sp}^2$  hybrid orbitals for bonding is \_\_\_\_\_.  
 $\text{NH}_3$ ,  $\text{SO}_2$ ,  $\text{SiO}_2$ ,  $\text{BeCl}_2$ ,  $\text{C}_2\text{H}_2$ ,  $\text{C}_2\text{H}_4$ ,  $\text{BCl}_3$ ,  $\text{HCHO}$ ,  $\text{C}_6\text{H}_6$ ,  $\text{BF}_3$ ,  $\text{C}_2\text{H}_4\text{Cl}_2$  **[JEE-Main-Online 2024]**
- 218.** Number of molecules from the following which are exceptions to octet rule is \_\_\_\_\_.  
 $\text{CO}_2$ ,  $\text{NO}_2$ ,  $\text{H}_2\text{SO}_4$ ,  $\text{BF}_3$ ,  $\text{CH}_4$ ,  $\text{SiF}_4$ ,  $\text{ClO}_2$ ,  $\text{PCl}_5$ ,  $\text{BeF}_2$ ,  $\text{C}_2\text{H}_6$ ,  $\text{CHCl}_3$ ,  $\text{CBr}_4$ .  
**[JEE-Main-Online 2024]**

- 219.** Arrange the following compounds in increasing order of their dipole moment :  
HBr,  $\text{H}_2\text{S}$ ,  $\text{NF}_3$  and  $\text{CHCl}_3$  [JEE-Main-Online 2025]  
(1)  $\text{NF}_3 < \text{HBr} < \text{H}_2\text{S} < \text{CHCl}_3$  (2)  $\text{HBr} < \text{H}_2\text{S} < \text{NF}_3 < \text{CHCl}_3$   
(3)  $\text{H}_2\text{S} < \text{HBr} < \text{NF}_3 < \text{CHCl}_3$  (4)  $\text{CHCl}_3 < \text{NF}_3 < \text{HBr} < \text{H}_2\text{S}$
- 220.** An element 'E' has the ionisation enthalpy value of  $374 \text{ kJ mol}^{-1}$ . 'E' reacts with elements A, B, C and D with electron gain enthalpy values of  $-328$ ,  $-349$ ,  $-325$  and  $-295 \text{ kJ mol}^{-1}$ , respectively. The correct order of the products EA, EB, EC and ED in terms of ionic character is : [JEE-Main-Online 2025]  
(1)  $\text{EB} > \text{EA} > \text{EC} > \text{ED}$  (2)  $\text{ED} > \text{EC} > \text{EA} > \text{EB}$   
(3)  $\text{EA} > \text{EB} > \text{EC} > \text{ED}$  (4)  $\text{ED} > \text{EC} > \text{EB} > \text{EA}$
- 221.** The molecules having square pyramidal geometry are [JEE-Main-Online 2025]  
(1)  $\text{BrF}_5$  &  $\text{XeOF}_4$  (2)  $\text{SbF}_5$  &  $\text{XeOF}_4$  (3)  $\text{SbF}_5$  &  $\text{PCl}_5$  (4)  $\text{BrF}_5$  &  $\text{PCl}_5$
- 222.** The type of oxide formed by the element among Li, Na, Be, Mg, B and Al that has the least atomic radius is : [JEE-Main-Online 2025]  
(1)  $\text{A}_2\text{O}_3$  (2)  $\text{AO}_2$  (3)  $\text{AO}$  (4)  $\text{A}_2\text{O}$
- 223.** Given below are two statements :  
**Statement (I)** : Experimentally determined oxygen-oxygen bond lengths in the  $\text{O}_3$  are found to be same and the bond length is greater than that of a  $\text{O}=\text{O}$  (double bond) but less than that of a single ( $\text{O}-\text{O}$ ) bond.  
**Statement (II)** : The strong lone pair-lone pair repulsion between oxygen atoms is solely responsible for the fact that the bond length in ozone is smaller than that of a double bond ( $\text{O}=\text{O}$ ) but more than that of a single bond ( $\text{O}-\text{O}$ ). [JEE-Main-Online 2025]  
In the light of the above statements, choose the **correct** answer from the options given below :  
(1) Statement I is true but Statement II is false (2) Both Statement I and Statement II are true  
(3) Both Statement I and Statement II are false (4) Statement I is false but Statement II is true
- 224.** Which of the following statement is true with respect to  $\text{H}_2\text{O}$ ,  $\text{NH}_3$  and  $\text{CH}_4$ ?  
(A) The central atoms of all the molecules are  $\text{sp}^3$  hybridized.  
(B) The  $\text{H}-\text{O}-\text{H}$ ,  $\text{H}-\text{N}-\text{H}$  and  $\text{H}-\text{C}-\text{H}$  angles in the above molecules are  $104.5^\circ$ ,  $107.5^\circ$  and  $109.5^\circ$  respectively.  
(C) The increasing order of dipole moment is  $\text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$ .  
(D) Both  $\text{H}_2\text{O}$  and  $\text{NH}_3$  are Lewis acids and  $\text{CH}_4$  is a Lewis base  
(E) A solution of  $\text{NH}_3$  in  $\text{H}_2\text{O}$  is basic. In this solution  $\text{NH}_3$  and  $\text{H}_2\text{O}$  act as Lowry-Bronsted acid and base respectively.  
Choose the correct answer from the options given below : [JEE-Main-Online 2025]  
(1) A, B and C only (2) C, D and E only (3) A, D and E only (4) A, B, C and E only
- 225.** Which of the following linear combination of atomic orbitals will lead to formation of molecular orbitals in homonuclear diatomic molecules [internuclear axis in z-direction]? [JEE-Main-Online 2025]  
(A)  $2p_z$  and  $2p_x$  (B)  $2s$  and  $2p_x$  (C)  $3d_{xy}$  and  $3d_{x^2-y^2}$  (D)  $2s$  and  $2p_z$   
(E)  $2p_z$  and  $3d_{x^2-y^2}$   
(1) E Only (2) A and B Only (3) D Only (4) C and D Only

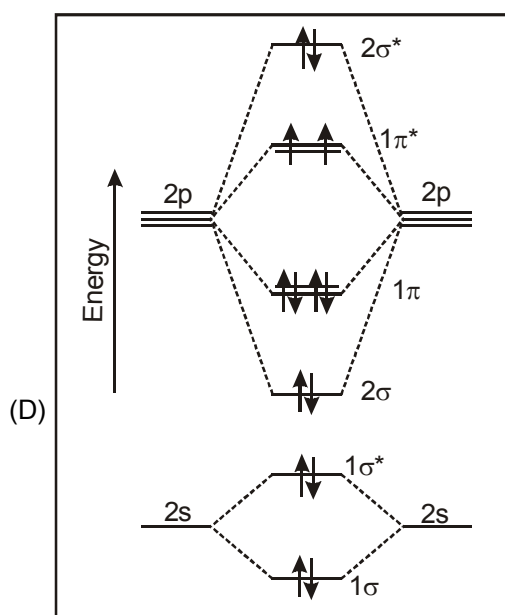
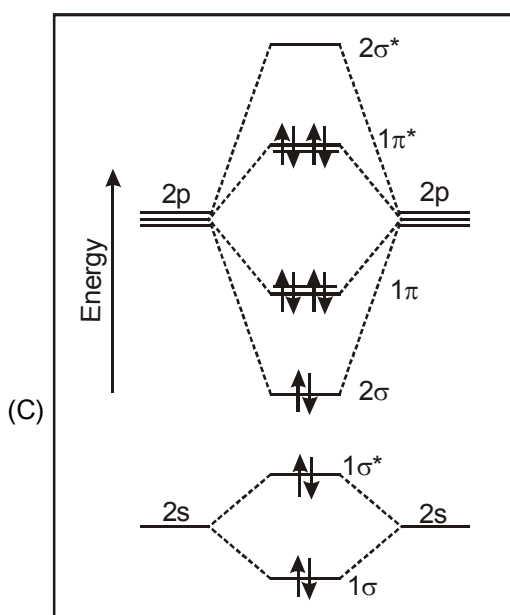
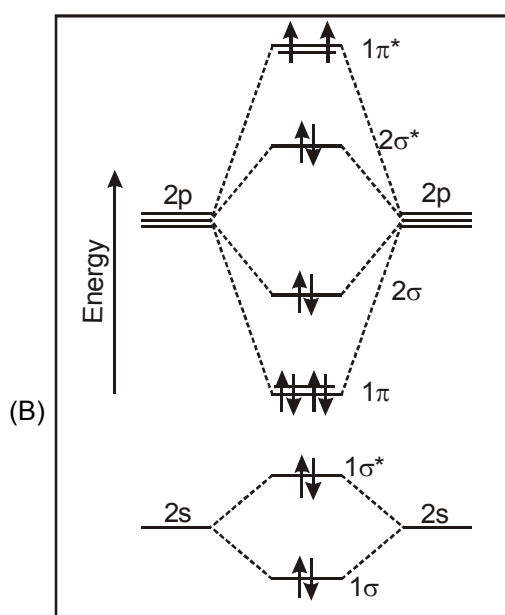
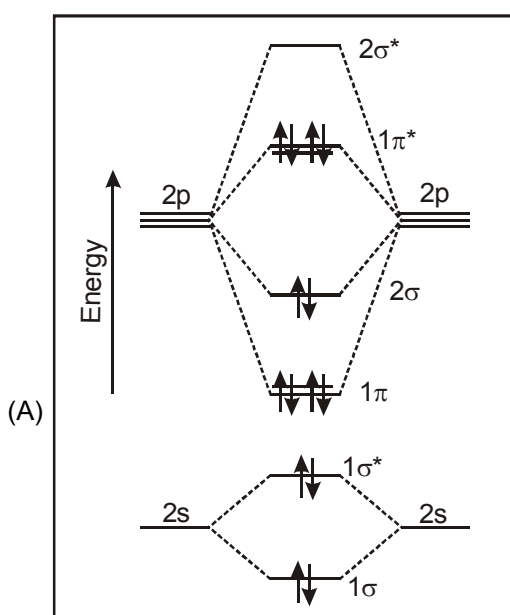
- 226.** The maximum covalency of a non-metallic group 15 element 'E' with weakest E–E bond is :  
**[JEE-Main-Online 2025]**  
 (1) 5 (2) 3 (3) 6 (4) 4
- 227.** Match the **LIST-I** with **LIST-II**  
**LIST-I**  
**(Classification of molecules based on octet rule)**  
 A. Molecules obeying octet rule  
 B. Molecules with incomplete octet  
 C. Molecules with incomplete octet with odd electron  
 D. Molecules with expanded octet  
 Choose the correct answer from the options given below :  
 (1) A-IV, B-II, C-I, D-III (2) A-III, B-II, C-I, D-IV  
 (3) A-IV, B-I, C-III, D-II (4) A-II, B-IV, C-III, D-I  
**LIST-II**  
**(Example)**  
 I. NO, NO<sub>2</sub>  
 II. BCl<sub>3</sub>, AlCl<sub>3</sub>  
 III. H<sub>2</sub>SO<sub>4</sub>, PCl<sub>5</sub>  
 IV. CCl<sub>4</sub>, CO<sub>2</sub>  
**[JEE-Main-Online 2025]**
- 228.** Total number of non bonded electrons present in NO<sub>2</sub><sup>-</sup> ion based on Lewis theory is \_\_\_\_\_.  
**[JEE-Main-Online 2025]**

# EXERCISE-IV

## PREVIOUS YEARS IIT-JEE QUESTIONS

Choose the correct alternative (only one correct answer).

- The increasing order of atomic radii of the following Group 13 elements is [JEE Advanced 2016]  
 (A)  $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$  (B)  $\text{Ga} < \text{Al} < \text{In} < \text{Tl}$   
 (C)  $\text{Al} < \text{In} < \text{Ga} < \text{Tl}$  (D)  $\text{Al} < \text{Ga} < \text{Tl} < \text{In}$
- The order of the oxidation state of the phosphorus atom in  $\text{H}_3\text{PO}_2$ ,  $\text{H}_3\text{PO}_4$ ,  $\text{H}_3\text{PO}_3$  and  $\text{H}_4\text{P}_2\text{O}_6$  is : [JEE Advanced 2017]  
 (A)  $\text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2$  (B)  $\text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6$   
 (C)  $\text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_4$  (D)  $\text{H}_3\text{PO}_4 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6$
- The correct molecular orbital diagram for  $\text{F}_2$  molecule in the ground state is [JEE Advanced 2023]



4. Based on **VSEPR** model, match the xenon compounds given in **List-I** with the corresponding geometries and the number of lone pairs on xenon given in **List-II** and choose the correct option.

**List-I**

- (P)  $\text{XeF}_2$   
(Q)  $\text{XeF}_4$   
(R)  $\text{XeO}_3$   
(S)  $\text{XeO}_3\text{F}_2$

**List-II**

- (1) Trigonal bipyramidal and two lone pair of electrons  
(2) Tetrahedral and one lone pair of electrons  
(3) Octahedral and two lone pair of electrons  
(4) Trigonal bipyramidal and no lone pair of electrons  
(5) Trigonal bipyramidal and three lone pair of electrons

[JEE Advanced 2024]

- (A)  $P \rightarrow 5, Q \rightarrow 2, R \rightarrow 3, S \rightarrow 1$   
(C)  $P \rightarrow 4, Q \rightarrow 3, R \rightarrow 2, S \rightarrow 1$

- (B)  $P \rightarrow 5, Q \rightarrow 3, R \rightarrow 2, S \rightarrow 4$   
(D)  $P \rightarrow 4, Q \rightarrow 2, R \rightarrow 5, S \rightarrow 3$

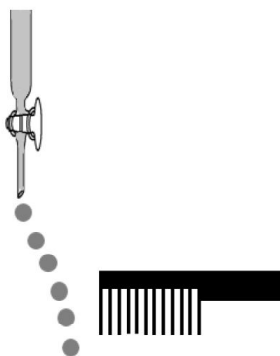
**One or more than one correct :**

5. The crystalline form of borax has [JEE Advanced 2016]  
(A) Tetranuclear  $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$  unit  
(B) All boron atoms in the same plane.  
(C) Equal number of  $\text{sp}^2$  and  $\text{sp}^3$  hybridized boron atoms.  
(D) One terminal hydroxide per boron atom.
6. The compound(s) with TWO lone pairs of electrons on the central atom is(are) [JEE Advanced 2016]  
(A)  $\text{BrF}_6$  (B)  $\text{ClF}_3$  (C)  $\text{XeF}_4$  (D)  $\text{SF}_4$
7. According to Molecular Orbital Theory, [JEE Advanced 2016]  
(A)  $\text{C}_2^{2-}$  is expected to be diamagnetic  
(B)  $\text{O}_2^{2+}$  is expected to have a longer bond length than  $\text{O}_2$   
(C)  $\text{N}_2^+$  and  $\text{N}_2^-$  have the same bond order  
(D)  $\text{He}_2^+$  has the same energy as two isolated He atoms
8. The colour of the  $\text{X}_2$  molecules of group 17 elements changes gradually from yellow to violet down the group. This is due to [JEE Advanced 2017]  
(A) decrease in HOMO-LUMO gap down the group  
(B) decrease in  $\pi^* - \sigma^*$  gap down the group  
(C) the physical state of  $\text{X}_2$  at room temperature changes from gas to solid down the group  
(D) decrease in ionization energy down the group
9. The correct statement(s) about the oxoacids,  $\text{HClO}_4$  and  $\text{HClO}$ , is(are) [JEE Advanced 2017]  
(A) The conjugate base of  $\text{HClO}_4$  is weaker base than  $\text{H}_2\text{O}$   
(B) The central atom in both  $\text{HClO}_4$  and  $\text{HClO}$  is  $\text{sp}^3$  hybridized  
(C)  $\text{HClO}_4$  is formed in the reaction between  $\text{Cl}_2$  and  $\text{H}_2\text{O}$   
(D)  $\text{HClO}_4$  is more acidic than  $\text{HClO}$  because of the resonance stabilization of its anion
10. The option(s) with only amphoteric oxides is(are) [JEE Advanced 2017]  
(A)  $\text{Cr}_2\text{O}_3, \text{CrO}, \text{SnO}, \text{PbO}$  (B)  $\text{NO}, \text{B}_2\text{O}_3, \text{PbO}, \text{SnO}_2$   
(C)  $\text{Cr}_2\text{O}_3, \text{BeO}, \text{SnO}, \text{SnO}_2$  (D)  $\text{ZnO}, \text{Al}_2\text{O}_3, \text{PbO}, \text{PbO}_2$
11. Among the following, the correct statement(s) is(are) [JEE Advanced 2017]  
(A)  $\text{Al}(\text{CH}_3)_3$  has the three-centre two-electron bonds in its dimeric structure  
(B)  $\text{AlCl}_3$  has the three-centre two-electron bonds in its dimeric structure  
(C)  $\text{BH}_3$  has the three-centre two-electron bonds in its dimeric structure  
(D) The Lewis acidity of  $\text{BCl}_3$  is greater than that of  $\text{AlCl}_3$

12. Based on the compounds of group 15 elements, the correct statement(s) is (are) **[JEE Advanced 2018]**  
 (A)  $\text{Bi}_2\text{O}_5$  is more basic than  $\text{N}_2\text{O}_5$   
 (B)  $\text{NF}_3$  is more covalent than  $\text{BiF}_3$   
 (C)  $\text{PH}_3$  boils at lower temperature than  $\text{NH}_3$   
 (D) The N–N single bond is stronger than the P–P single bond
13. The compound(s) which generate(s)  $\text{N}_2$  gas upon thermal decomposition below  $300^\circ\text{C}$  is(are) **[JEE Advanced 2018]**  
 (A)  $\text{NH}_4\text{NO}_3$  (B)  $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$  (C)  $\text{Ba}(\text{N}_3)_2$  (D)  $\text{Mg}_3\text{N}_2$
14. Each of the following options contains a set of four molecules. Identify the option(s) where all four molecules possess permanent dipole moment at room temperature. **[JEE Advanced 2019]**  
 (A)  $\text{SO}_2$ ,  $\text{C}_6\text{H}_5\text{Cl}$ ,  $\text{H}_2\text{Se}$ ,  $\text{BrF}_5$  (B)  $\text{BeCl}_2$ ,  $\text{CO}_2$ ,  $\text{BCl}_3$ ,  $\text{CHCl}_3$   
 (C)  $\text{NO}_2$ ,  $\text{NH}_3$ ,  $\text{POCl}_3$ ,  $\text{CH}_3\text{Cl}$  (D)  $\text{BF}_3$ ,  $\text{O}_3$ ,  $\text{SF}_6$ ,  $\text{XeF}_6$
15. For diatomic molecules, the correct statement(s) about the molecular orbitals formed by the overlap of two  $2p_z$  orbitals is(are) **[JEE Advanced 2022]**  
 (A)  $\sigma$  orbital has a total of two nodal planes.  
 (B)  $\sigma^*$  orbital has one node in the xz-plane containing the molecular axis.  
 (C)  $\pi$  orbital has one node in the plane which is perpendicular to the molecular axis and goes through the center of the molecule.  
 (D)  $\pi^*$  orbital has one node in the xy-plane containing the molecular axis.
16. The compound(s) having peroxide linkage is(are) **[JEE Advanced 2024]**  
 (A)  $\text{H}_2\text{S}_2\text{O}_7$  (B)  $\text{H}_2\text{S}_2\text{O}_8$  (C)  $\text{H}_2\text{S}_2\text{O}_5$  (D)  $\text{H}_2\text{SO}_5$
17. The option(s) in which at least three molecules follow Octet Rule is(are) **[JEE Advanced 2024]**  
 (A)  $\text{CO}_2$ ,  $\text{C}_2\text{H}_4$ ,  $\text{NO}$  and  $\text{HCl}$  (B)  $\text{NO}_2$ ,  $\text{O}_3$ ,  $\text{HCl}$  and  $\text{H}_2\text{SO}_4$   
 (C)  $\text{BCl}_3$ ,  $\text{NO}$ ,  $\text{NO}_2$  and  $\text{H}_2\text{SO}_4$  (D)  $\text{CO}_2$ ,  $\text{BCl}_3$ ,  $\text{O}_3$  and  $\text{C}_2\text{H}_4$
18. The correct statement(s) about intermolecular forces is(are) **[JEE Advanced 2025]**  
 (A) The potential energy between two point charges approaches zero more rapidly than the potential energy between a point dipole and a point charge as the distance between them approaches infinity.  
 (B) The average potential energy of two rotating polar molecules that are separated by a distance  $r$  has  $1/r^3$  dependence.  
 (C) The dipole-induced dipole average interaction energy is independent of temperature.  
 (D) Nonpolar molecules attract one another even though neither has a permanent dipole moment.
19. Regarding the molecular orbital (MO) energy levels for homonuclear diatomic molecules, the **INCORRECT** statement(s) is(are) **[JEE Advanced 2025]**  
 (A) Bond order of  $\text{Ne}_2$  is zero.  
 (B) The highest occupied molecular orbital (HOMO) of  $\text{F}_2$  is  $\sigma$ -type.  
 (C) Bond energy of  $\text{O}_2^+$  is smaller than the bond energy of  $\text{O}_2$ .  
 (D) Bond length of  $\text{Li}_2$  is larger than the bond length of  $\text{B}_2$ .

**Integers Type :**

20. The sum of the number of lone pairs of electrons on each central atom in the following species is :  
[TeBr<sub>6</sub>]<sup>2-</sup>, [BrF<sub>2</sub>]<sup>+</sup>, SNF<sub>3</sub>, and [XeF<sub>3</sub>]<sup>-</sup>  
(Atomic numbers : N = 7, F = 9, S = 16, Br = 35, Te = 52, Xe = 54) [JEE Advanced 2017]
21. Among H<sub>2</sub>, He<sub>2</sub><sup>+</sup>, Li<sub>2</sub>, Be<sub>2</sub>, B<sub>2</sub>, C<sub>2</sub>, N<sub>2</sub>, O<sub>2</sub><sup>-</sup>, and F<sub>2</sub>, the number of diamagnetic species is  
(Atomic numbers : H = 1, He = 2, Li = 3, Be = 4, B = 5, C = 6, N = 7, O = 8, F = 9) [JEE Advanced 2017]
22. The total number of compounds having at least one bridging oxo group among the molecules given below is \_\_\_\_\_.  
N<sub>2</sub>O<sub>3</sub>, N<sub>2</sub>O<sub>5</sub>, P<sub>4</sub>O<sub>6</sub>, P<sub>4</sub>O<sub>7</sub>, H<sub>4</sub>P<sub>2</sub>O<sub>5</sub>, H<sub>5</sub>P<sub>3</sub>O<sub>10</sub>, H<sub>2</sub>S<sub>2</sub>O<sub>3</sub>, H<sub>2</sub>S<sub>2</sub>O<sub>5</sub> [JEE Advanced 2018]
23. Among B<sub>2</sub>H<sub>6</sub>, B<sub>3</sub>N<sub>3</sub>H<sub>6</sub>, N<sub>2</sub>O, N<sub>2</sub>O<sub>4</sub>, H<sub>2</sub>S<sub>2</sub>O<sub>3</sub> and H<sub>2</sub>S<sub>2</sub>O<sub>8</sub>, the total number of molecules containing covalent bond between two atoms of the same kind is \_\_\_\_\_. [JEE Advanced 2019]
24. Consider the following compounds in the liquid form :  
O<sub>2</sub>, HF, H<sub>2</sub>O, NH<sub>3</sub>, H<sub>2</sub>O<sub>2</sub>, CCl<sub>4</sub>, CHCl<sub>3</sub>, C<sub>6</sub>H<sub>6</sub>, C<sub>6</sub>H<sub>5</sub>Cl.  
When a charged comb is brought near their flowing stream, how many of them show deflection as per the following figure ? [JEE Advanced 2020]



25. Among [I<sub>3</sub>]<sup>+</sup>, [SiO<sub>4</sub>]<sup>4-</sup>, SO<sub>2</sub>Cl<sub>2</sub>, XeF<sub>2</sub>, SF<sub>4</sub>, ClF<sub>3</sub>, Ni(CO)<sub>4</sub>, XeO<sub>2</sub>F<sub>2</sub>, [PtCl<sub>4</sub>]<sup>2-</sup>, XeF<sub>4</sub>, and SOCl<sub>2</sub>, the total number of species having sp<sup>3</sup> hybridised central atom is \_\_\_\_\_. [JEE Advanced 2023]
26. Consider the following molecules: Br<sub>3</sub>O<sub>8</sub>, F<sub>2</sub>O, H<sub>2</sub>S<sub>4</sub>O<sub>6</sub>, H<sub>2</sub>S<sub>5</sub>O<sub>6</sub>, and C<sub>3</sub>O<sub>2</sub>.  
Count the number of atoms existing in their zero oxidation state in each molecule.  
Their sum is \_\_\_\_\_. [JEE Advanced 2023]

# ANSWER KEY

## EXERCISE-I

|      |   |      |   |      |   |      |   |      |   |      |    |      |   |
|------|---|------|---|------|---|------|---|------|---|------|----|------|---|
| 1.   | 1 | 2.   | 2 | 3.   | 3 | 4.   | 3 | 5.   | 1 | 6.   | 2  | 7.   | 3 |
| 8.   | 1 | 9.   | 1 | 10.  | 4 | 11.  | 3 | 12.  | 4 | 13.  | 4  | 14.  | 3 |
| 15.  | 3 | 16.  | 4 | 17.  | 3 | 18.  | 1 | 19.  | 2 | 20.  | 3  | 21.  | 1 |
| 22.  | 4 | 23.  | 3 | 24.  | 4 | 25.  | 2 | 26.  | 2 | 27.  | 4  | 28.  | 2 |
| 29.  | 4 | 30.  | 1 | 31.  | 3 | 32.  | 4 | 33.  | 3 | 34.  | 2  | 35.  | 3 |
| 36.  | 3 | 37.  | 3 | 38.  | 1 | 39.  | 2 | 40.  | 4 | 41.  | 1  | 42.  | 1 |
| 43.  | 4 | 44.  | 2 | 45.  | 2 | 46.  | 2 | 47.  | 1 | 48.  | 3  | 49.  | 2 |
| 50.  | 2 | 51.  | 2 | 52.  | 3 | 53.  | 3 | 54.  | 2 | 55.  | 2  | 56.  | 3 |
| 57.  | 4 | 58.  | 3 | 59.  | 2 | 60.  | 4 | 61.  | 4 | 62.  | 3  | 63.  | 3 |
| 64.  | 3 | 65.  | 3 | 66.  | 2 | 67.  | 4 | 68.  | 1 | 69.  | 3  | 70.  | 3 |
| 71.  | 4 | 72.  | 2 | 73.  | 2 | 74.  | 3 | 75.  | 2 | 76.  | 2  | 77.  | 2 |
| 78.  | 1 | 79.  | 3 | 80.  | 1 | 81.  | 3 | 82.  | 3 | 83.  | 2  | 84.  | 1 |
| 85.  | 2 | 86.  | 4 | 87.  | 3 | 88.  | 2 | 89.  | 1 | 90.  | 3  | 91.  | 3 |
| 92.  | 4 | 93.  | 2 | 94.  | 1 | 95.  | 1 | 96.  | 3 | 97.  | 2  | 98.  | 1 |
| 99.  | 4 | 100. | 1 | 101. | 3 | 102. | 2 | 103. | 4 | 104. | 1  | 105. | 2 |
| 106. | 2 | 107. | 3 | 108. | 4 | 109. | 2 | 110. | 1 | 111. | 3  | 112. | 1 |
| 113. | 3 | 114. | 1 | 115. | 3 | 116. | 3 | 117. | 5 | 118. | 4  | 119. | 4 |
| 120. | 0 | 121. | 4 | 122. | 3 | 123. | 4 | 124. | 3 | 125. | 13 | 126. | 4 |

## EXERCISE-II

|     |   |     |   |     |   |     |   |     |   |     |   |     |   |
|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|
| 1.  | 4 | 2.  | 1 | 3.  | 3 | 4.  | 4 | 5.  | 4 | 6.  | 1 | 7.  | 1 |
| 8.  | 3 | 9.  | 2 | 10. | 2 | 11. | 1 | 12. | 4 | 13. | 2 | 14. | 4 |
| 15. | 3 | 16. | 2 | 17. | 4 | 18. | 2 | 19. | 3 | 20. | 3 | 21. | 2 |
| 22. | 1 | 23. | 3 | 24. | 3 | 25. | 3 | 26. | 3 | 27. | 1 | 28. | 1 |
| 29. | 1 | 30. | 3 | 31. | 4 | 32. | 1 | 33. | 1 | 34. | 3 | 35. | 2 |
| 36. | 3 | 37. | 2 | 38. | 2 | 39. | 3 | 40. | 1 | 41. | 2 | 42. | 2 |
| 43. | 4 | 44. | 4 | 45. | 1 | 46. | 1 | 47. | 3 | 48. | 4 | 49. | 3 |
| 50. | 3 | 51. | 3 | 52. | 1 | 53. | 1 | 54. | 3 | 55. | 2 | 56. | 4 |
| 57. | 3 | 58. | 2 | 59. | 4 | 60. | 3 |     |   |     |   |     |   |

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## EXERCISE-III

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|      |       |      |       |      |       |      |        |      |       |      |       |      |       |
|------|-------|------|-------|------|-------|------|--------|------|-------|------|-------|------|-------|
| 1.   | 1     | 2.   | 1     | 3.   | 4     | 4.   | 2      | 5.   | 4     | 6.   | 4     | 7.   | 3     |
| 8.   | 4     | 9.   | 1     | 10.  | 2     | 11.  | 4      | 12.  | 4     | 13.  | 1     | 14.  | 4     |
| 15.  | 19.00 | 16.  | 3.00  | 17.  | 1     | 18.  | 3      | 19.  | 4     | 20.  | 3     | 21.  | 2     |
| 22.  | 3     | 23.  | 4     | 24.  | 3     | 25.  | 2      | 26.  | 2     | 27.  | 4     | 28.  | 4     |
| 29.  | 4     | 30.  | 3     | 31.  | 2     | 32.  | 1      | 33.  | 4     | 34.  | 3     | 35.  | 4     |
| 36.  | 2     | 37.  | 2     | 38.  | 3     | 39.  | 3      | 40.  | 4     | 41.  | 1     | 42.  | 2     |
| 43.  | 2     | 44.  | 2     | 45.  | 4     | 46.  | 2      | 47.  | 3     | 48.  | 3     | 49.  | 1     |
| 50.  | 3     | 51.  | 2     | 52.  | 4     | 53.  | 173.00 | 54.  | 2.00  | 55.  | 03.00 | 56.  | 00.00 |
| 57.  | 0.00  | 58.  | 10.00 | 59.  | 3.00  | 60.  | 10.00  | 61.  | 2.00  | 62.  | 3     | 63.  | 2     |
| 64.  | 2     | 65.  | 1     | 66.  | 1     | 67.  | 3      | 68.  | 3     | 69.  | 2     | 70.  | 3     |
| 71.  | 3     | 72.  | 2     | 73.  | 3     | 74.  | 1      | 75.  | 2     | 76.  | 2     | 77.  | 1     |
| 78.  | 3     | 79.  | 1     | 80.  | 3     | 81.  | 3      | 82.  | 1     | 83.  | 1     | 84.  | 4     |
| 85.  | 2     | 86.  | 2     | 87.  | 1     | 88.  | 3      | 89.  | 3     | 90.  | 4     | 91.  | 1     |
| 92.  | 3     | 93.  | 1     | 94.  | 2     | 95.  | 1      | 96.  | 3     | 97.  | 2     | 98.  | 4     |
| 99.  | 2     | 100. | 2     | 101. | 2     | 102. | 3.00   | 103. | 3.00  | 104. | 2.00  | 105. | 4.00  |
| 106. | 1.00  | 107. | 4.00  | 108. | 4.00  | 109. | 3.00   | 110. | 6.00  | 111. | 3.00  | 112. | 1.00  |
| 113. | 0.00  | 114. | 2.00  | 115. | 3     | 116. | 3      | 117. | 3     | 118. | 2     | 119. | 1     |
| 120. | 3     | 121. | 3     | 122. | 1     | 123. | 4      | 124. | 3     | 125. | 1     | 126. | 2     |
| 127. | 2     | 128. | 3     | 129. | 4     | 130. | 4      | 131. | 3     | 132. | 3     | 133. | 3     |
| 134. | 1     | 135. | 3     | 136. | 3     | 137. | 2      | 138. | 4     | 139. | 1     | 140. | 2     |
| 141. | 3     | 142. | 4     | 143. | 2     | 144. | 1      | 145. | 3     | 146. | 3     | 147. | 06.00 |
| 148. | 5.00  | 149. | 03.00 | 150. | 17.00 | 151. | 03.00  | 152. | 3.00  | 153. | 04.00 | 154. | 01.00 |
| 155. | 03.00 | 156. | 04.00 | 157. | 05.00 | 158. | 04.00  | 159. | 12.00 | 160. | 8.00  | 161. | 3.00  |
| 162. | 6.00  | 163. | 3.00  | 164. | 5.00  | 165. | 5.00   | 166. | 4.00  | 167. | 2     | 168. | 1     |
| 169. | 2     | 170. | 2     | 171. | 4     | 172. | 3      | 173. | 2     | 174. | 4     | 175. | 1     |
| 176. | 1     | 177. | 4     | 178. | 3     | 179. | 1      | 180. | 3.00  | 181. | 4.00  | 182. | 4.00  |
| 183. | 8.00  | 184. | 1.00  | 184. | 4.00  | 186. | 6.00   | 187. | 3.00  | 188. | 4.00  | 189. | 6.00  |
| 190. | 0.00  | 191. | 2     | 192. | 3     | 193. | 3      | 194. | 3     | 195. | 1     | 196. | 2     |
| 197. | 3     | 198. | 3     | 199. | 1     | 200. | 1      | 201. | 4     | 202. | 4     | 203. | 4     |
| 204. | 4     | 205. | 4     | 206. | 1     | 207. | 1      | 208. | 2     | 209. | 2.00  | 210. | 4.00  |
| 211. | 6.00  | 212. | 2.00  | 213. | 5.00  | 214. | 8.00   | 215. | 6.00  | 216. | 5.00  | 217. | 6.00  |
| 218. | 6.00  | 219. | 1     | 220. | 3     | 221. | 1      | 222. | 1     | 223. | 1     | 224. | 1     |
| 225. | 3     | 226. | 4     | 227. | 1     | 228. | 12.00  |      |       |      |       |      |       |

## EXERCISE-IV

Choose the correct alternative (only one correct answer) :

1. B      2. A      3. C      4. B

One or more than one correct :

5. A,C,D    6. B,C    7. A,C    8. AB    9. ABD    10. C,D    11. A,C,D  
12. A,B,C    13. B,C    14. A, C    15. A, D    16. B,D    17. A,D    18. B, D  
19. B, C

Integers Type :

20. 6      21. 6      22. 5 or 6    23. 4      24. 6      25. 5.00    26. 6.00